

Logic

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September 14, 2026

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Chapter 1

An Introduction to Mathematics

1.1 Informal Logic

Definition 1.1.1. A *proposition* is the meaning of a declarative sentence.

Definition 1.1.2. An *argument* is a collection of propositions called *premises* and a single proposition called the *conclusion*.

Definition 1.1.3. An argument is *valid* if, whenever all its premises are true, the conclusion must also be true.

Definition 1.1.4. An argument is *sound* if it is valid and all its premises are actually true.

Example 1.1.5. Consider the classic argument:

1. All humans are mortal. (Premise)
2. Socrates is a human. (Premise)
3. Therefore, Socrates is mortal. (Conclusion)

This argument is valid and sound.

Example 1.1.6. 1. All birds can fly.

2. Penguins are birds.
3. Therefore penguins can fly.

This argument is valid (its structure is identical to that of Example 1.1.5) but it is not sound.

Exercise 1.1. Show that the conclusion of a sound argument is true.

When our informal reasoning goes awry, we run into paradoxes. These show that naive language can easily contradict itself.

Exercise 1.2. We assume that a person is either a perpetual liar (who never speaks the truth) or completely truthful (who never lies). Adam says about himself and Bill: “Both of us are liars”. Determine who is a liar and who is truthful.

Exercise 1.3. Epimenides of Knossos was a Cretan philosopher. He once said “Cretans, liars always, evil beasts, idle bellies”. Logically, this sentence is unproblematic, but what happens if we make the same assumption as in exercise 1.2?

Remark 1.1.7. The paradox in the above exercise is known as the Epimenides paradox. It will become quite clear later, that this, very simple paradox leads to incredibly sophisticated results about logic itself.

Exercise 1.4. A number $1 < k < 9$ is chosen. Andy is told the value of the greatest prime that divides k and Belle is told the value of the remainder of k when dividing by 5. Belle tells Andy “I don’t know k ”. and Andy replies “Initially, I didn’t know either, but now I do!”. What is k ?

1.2 Naive Set Theory

We require a basic language of set theory to articulate clearly the concepts used in logic.

1.2.1 A Logical Shorthand Primer

Defining concepts of set theory cleanly requires logical notation. Let φ and ψ be formulae. The meaning of these formulae will be defined later:

Symbol	Name	Read as...
$\top$	Verum	“Tautology”
$\perp$	Falsum	“Contradiction”
$\neg\varphi$	Negation	“Not φ ”
$\varphi \wedge \psi$	Conjunction	“ φ and ψ ”
$\varphi \vee \psi$	Disjunction	“ φ or ψ ”
$\varphi \rightarrow \psi$	Conditional (Implication)	“If φ , then ψ ”
$\varphi \iff \psi$	Biconditional (Equivalence)	“ φ if and only if ψ ”
$\forall x \varphi$	Universal Quantifier	“For every x , φ ”
$\exists x \varphi$	Existential Quantifier	“For some x , φ ”

Definition 1.2.1. A *set* is a collection of objects.

Definition 1.2.2. Let A be a set. The statement $a \in A$ is read as “ a is an element of A ” and is a logical sentence.

Definition 1.2.3. A set A is a *subset* of B , written $A \subseteq B$, if $\forall x (x \in A \rightarrow x \in B)$. We also call B a *superset* of A .

Remark 1.2.4. We introduce the bounded quantifier shorthand: $\forall x \in A \psi$ is written as an abbreviation for $\forall x (x \in A \rightarrow \psi)$, and $\exists x \in A \psi$ is written for $\exists x (x \in A \wedge \psi)$.

Definition 1.2.5. Equality for sets is defined extensionally: $A = B \iff \forall x (x \in A \iff x \in B)$.

Exercise 1.5. Prove that $A = B \iff A \subseteq B \wedge B \subseteq A$.

Exercise 1.6. Prove that $\forall A \ A \subseteq A$.

Exercise 1.7. Prove that $\forall A \ \emptyset \subseteq A$.

Definition 1.2.6. $A := \{x \mid \varphi(x)\}$ is the set defined by $\forall x (x \in A \iff \varphi(x))$.

Definition 1.2.7. $\{x \in X \mid \varphi(x)\} := \{x \mid x \in X \wedge \varphi(x)\}$

Definition 1.2.8. $\emptyset := \{x \mid \perp\}$

Definition 1.2.9. An *ordered pair* (a, b) is a mathematical object satisfying the fundamental property: $(a, b) = (c, d) \iff a = c \wedge b = d$.

Remark 1.2.10. Note that Definition 1.2.9 gives only the property of an ordered pair, not its set-theoretic definition. We shall return to this point in Chapter 7.

Definition 1.2.11. The *Cartesian product* $A \times B$ is defined as: $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.

Definition 1.2.12. For any set A , we define: $A^0 := \{\emptyset\}$, and $A^{n+1} := A^n \times A$.

Exercise 1.8. Why is it wrong to say that $A^n = A \times \dots \times A$ (n times)?

Hint: Try an example with $n = 3$.

Definition 1.2.13. In light of Exercise 1.8, we take the operation $\times$ to be left-associative. This is an ad hoc convention that will be refined later.

Definition 1.2.14. The *difference* of two sets A and B is defined as: $A \setminus B := \{a \in A \mid a \notin B\}$.

Definition 1.2.15. The *power set* of a set A is defined as: $\mathcal{P}(A) := \{x \mid x \subseteq A\}$.

Exercise 1.9. Prove that if $A \subseteq B$ for two sets A, B , then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Definition 1.2.16. Let X be a designated set called the *universe set*. For any set $A \subseteq X$, we define its *complement* relative to X as: $A^c := X \setminus A$.

Exercise 1.10. Prove that if $A \subseteq B$, then $B^c \subseteq A^c$ for a fixed universe set.

Exercise 1.11. Prove that $(A^c)^c = A$.

Definition 1.2.17. The *binary union* is defined as: $A \cup B := \{x \mid x \in A \vee x \in B\}$.

Definition 1.2.18. The general *union* of a set of sets A is defined as: $\bigcup A := \{x \mid \exists y \in A \ x \in y\}$.

Exercise 1.12. Prove that $A \cup B = B \cup A$.

Exercise 1.13. Prove that $A \subseteq A \cup B$.

Exercise 1.14. Prove that $B \subseteq A \cup B$ without repeating the calculation from Exercise 1.13.

Exercise 1.15. Prove that $(A \cup B) \cup C = A \cup (B \cup C)$.

Remark 1.2.19. Exercise 1.15 shows that the expression “ $A \cup B \cup C$ ” is unambiguous and mathematically valid.

Definition 1.2.20. The *binary intersection* is defined as: $A \cap B := \{x \mid x \in A \wedge x \in B\}$.

Definition 1.2.21. For $A \neq \emptyset$, the general *intersection* is: $\bigcap A := \{x \mid \forall y \in A \ x \in y\}$.

Definition 1.2.22. Let $\mathcal{A}$ be a set of sets. We define $\bigcup_{A \in \mathcal{A}} A := \bigcup \mathcal{A}$, similarly for $\bigcap_{A \in \mathcal{A}} A$ if $\mathcal{A} \neq \emptyset$.

Remark 1.2.23. When considering a designated universe set X , we define: $\bigcap A := \{x \in X \mid \forall y \in A \ x \in y\}$.

Exercise 1.16. Prove that for any $A \neq \emptyset$, the definitions of $\bigcap A$ given in 1.2.21 and 1.2.23 agree.

Exercise 1.17. Prove that $(A \cup B)^c = A^c \cap B^c$ for a fixed universe.

Exercise 1.18. Prove that $(A \cap B) \cap C = A \cap (B \cap C)$ without repeating the dual calculation.

Exercise 1.19. $(\star)$ Prove that $(A \cup B) \times C = (A \times C) \cup (B \times C)$.

Exercise 1.20. Prove that $A \setminus B = A \cap B^c$ for a fixed universe set.

1.3 Representations of Relations and Functions

Here we try to formalise relations and functions as sets.

Definition 1.3.1. A set representation of an n -ary relation is a subset $R \subseteq A_1 \times \dots \times A_n$.

Remark 1.3.2. This representation carries a subtlety: nothing intrinsic to a set $R \subseteq A \times B$ tells us whether it represents a binary relation on A and B , or a unary relation on the set $A \times B$. We could resolve this by formally defining an n -ary relation as a pair (R, n) , but it is simpler to just keep the ambiguity in mind.

Remark 1.3.3. We frequently conflate a relation with its set representation. For example, if R is a binary relation and R^V is its set representation, we write $R^V(x, y)$ to denote $(x, y) \in R^V$.

Definition 1.3.4. For a relation $R \subseteq A_1 \times \dots \times A_n \times B$, we define its *domain* as: $\text{dom } R := \{a \mid \exists b \in B (a, b) \in R\}$, and its *range* as: $\text{ran } R := \{b \mid \exists a (a, b) \in R\}$.

Example 1.3.5. If $(w, x, y, z) \in R$, then $(w, x, y) \in \text{dom } R$ and $z \in \text{ran } R$.

Definition 1.3.6. A binary relation is any subset $R \subseteq A \times B$.

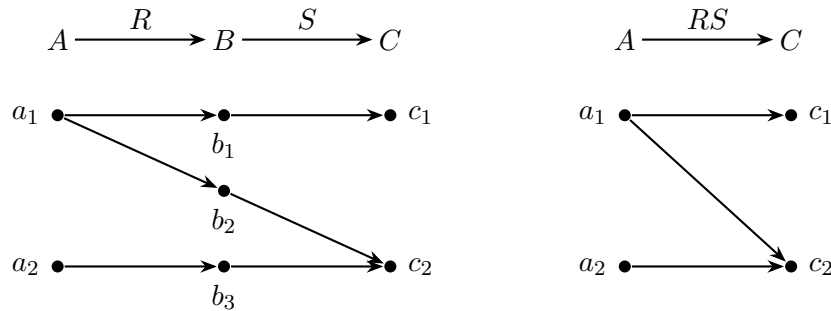
Definition 1.3.7. A binary relation $R \subseteq A \times B$ is *functional* if for every $a \in A$, there exists at most one $b \in B$ such that $(a, b) \in R$.

Definition 1.3.8. A (set representation of a) *function* $f: A \rightarrow B$ is any functional relation $f \subseteq A \times B$ such that $\text{dom } f = A$.

Definition 1.3.9. For a function $f: A \rightarrow B$ and $x \in A$, we write $f(x) := y$ where y is the unique element in B for which $(x, y) \in f$.

Remark 1.3.10. We encounter the same representational ambiguity highlighted in Remark 1.3.2. Let $f := \{((0, 0), 0), ((1, 0), 1)\} \subseteq \{0, 1\} \times \{0\} \times \{0, 1\}$. This set can be viewed as representing either a function $f: \{0, 1\} \rightarrow \{0\} \times \{0, 1\}$ or a function $f: \{0, 1\} \times \{0\} \rightarrow \{0, 1\}$.

Definition 1.3.11. For two binary relations $R \subseteq A \times B$ and $S \subseteq B \times C$, we define their *composition* as: $RS := \{(a, c) \in A \times C \mid \exists b ((a, b) \in R \wedge (b, c) \in S)\}$.

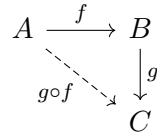


Two relations $R \subseteq A \times B$ and $S \subseteq B \times C$. The composite $RS \subseteq A \times C$.

Exercise 1.21. Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Prove that $(RS)T = R(ST)$.

Remark 1.3.12. Exercise 1.21 permits us to drop parentheses when composing relations, writing simply RST .

Definition 1.3.13. For two functions $f: A \rightarrow B$ and $g: B \rightarrow C$, the function $g \circ f: A \rightarrow C$ is defined as: $g \circ f := \{(a, c) \in A \times C \mid \exists b (f(a) = b \wedge g(b) = c)\}$.



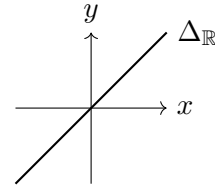
Exercise 1.22. Prove that $(g \circ f)(x) = g(f(x))$.

Exercise 1.23. Prove that $g \circ f = fg$ (which compares function composition with relational composition).

Remark 1.3.14. Note that function composition is conventionally written right-to-left ($g \circ f$), whereas relational composition is written left-to-right (fg). This is because we write the argument of a function on the right, but relational membership is evaluated left-to-right.

Definition 1.3.15. The *diagonal relation* on a set X is defined as $\Delta_X := \{(x, x) \mid x \in X\}$. When viewed as a function, we denote it by id_X and call it the *identity function* on X .

Remark 1.3.16. The term “diagonal relation” arises because when graphing this relation on a Cartesian coordinate plane, it forms a straight line oriented at a $\frac{\pi}{4}$ angle.



Definition 1.3.17. Given a binary relation $R \subseteq A \times B$, we define its *converse* (or *transpose*) relation as: $R^T := \{(b, a) \in B \times A \mid (a, b) \in R\}$.

Definition 1.3.18. A function $\iota : A \rightarrow B$ that satisfies $\forall a \in A \iota(a) = a$ is called an inclusion from A to B .

Definition 1.3.19. A function $f : A \rightarrow B$ is an *injection* (adj: *injective*) if: $\forall x \forall y (f(x) = f(y) \rightarrow x = y)$.

Exercise 1.24. Prove that an inclusion from A to B is injective.

Exercise 1.25. Prove that the composition of two injections is an injection.

Exercise 1.26. Prove that the composition of an inclusion from A to B with an inclusion from B to C (in the only possible way) is an inclusion from A to C .

Definition 1.3.20. A function $f : A \rightarrow B$ is a *surjection* (adj: *surjective*) if: $\forall y \in B \exists x \in A f(x) = y$.

Exercise 1.27. Prove that the composition of two surjections is a surjection.

Definition 1.3.21. A function $f : A \rightarrow B$ is a *bijection* (adj: *bijective*) if it is both injective and surjective.

Exercise 1.28. Conclude that the composition of two bijections is a bijection.

Definition 1.3.22. A function $g: B \rightarrow A$ is an *inverse* of a function $f: A \rightarrow B$ if $f \circ g = \text{id}_B$ and $g \circ f = \text{id}_A$. We then call f *invertible*.

Exercise 1.29. $(\star)$ Prove that a function is invertible if and only if it is bijective.

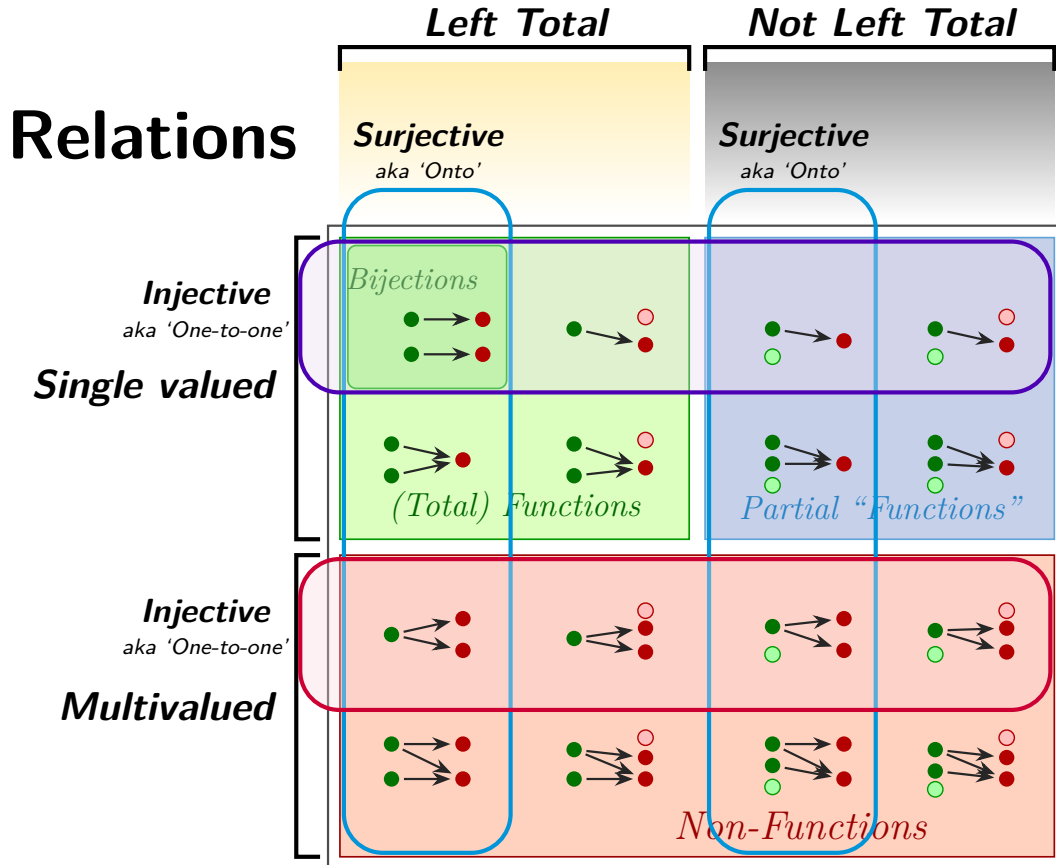


Figure 1.1: Classification of binary relations.

Definition 1.3.23. Let $f: A \rightarrow B$ be a function. The *image* of $X \subseteq A$ under f is defined as: $f(X) := \{b \in B \mid \exists x \in X, f(x) = b\}$.

Exercise 1.30. Let $f: A \rightarrow B$ and $X \subseteq A$. Prove that $f(X) = \{f(x) \mid x \in X\}$.

Definition 1.3.24. Let $f: A \rightarrow B$ be a function. The *preimage* of $Y \subseteq B$ under f is defined as: $f^{-1}(Y) := \{a \in A \mid f(a) \in Y\}$.

Exercise 1.31. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$. Prove that $(g \circ f)(A) = g(f(A))$.

Exercise 1.32. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$. Prove that $(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A))$.

Exercise 1.33. Consider three functions f, g, h such that their compositions make sense. Prove that $(f \circ g) \circ h = f \circ (g \circ h)$ without any calculation.

Definition 1.3.25. Let $f: A \rightarrow B$ be a function and $S \subseteq A$. The *restriction* of f to S , denoted $f|_S$, is defined as $f \cap (S \times B)$.

Exercise 1.34. Let $f: A \rightarrow B$ and $S \subseteq A$. Prove that $\forall x \in S \ f|_S(x) = f(x)$.

Definition 1.3.26. Let $R \subseteq A_1 \times \dots \times A_n$ be an n -ary relation and let $B_1 \times \dots \times B_n \subseteq A_1 \times \dots \times A_n$. The *restriction* of R to $B_1 \times \dots \times B_n$ is defined as: $R|_{B_1 \times \dots \times B_n} := R \cap (B_1 \times \dots \times B_n)$. If $R \subseteq A^n$ and $B \subseteq A$, we write $R|_B := R \cap B^n$.

Definition 1.3.27. Let A and B be sets. We define the set of functions from A to B as $B^A := \{f: A \rightarrow B\}$.

Definition 1.3.28. Let $f: A \rightarrow B$ be a function and for every $a \in A$, $f(a)$ be a set. We define $\prod_{a \in A} f(a) := \{g \in (\bigcup B)^A \mid \forall a \in A \ g(a) \in f(a)\}$.

Definition 1.3.29. The set of natural numbers is defined as $\mathbb{N} := \{0, 1, 2, \dots\}$.

Remark 1.3.30. The above definition of $\mathbb{N}$ is purely informal. We will formally redefine natural numbers set-theoretically in Chapter 7.

Definition 1.3.31. The set of integers is informally defined as $\mathbb{Z} := \{0, 1, -1, 2, -2, \dots\}$.

1.4 Equivalence Relations and Quotients

We now explore relations that behave like equality, which allow us to group objects together and abstract away their individual differences.

Definition 1.4.1. A binary relation $R \subseteq A \times A$ is said to be:

Property	Condition
Reflexive	$\forall x \in A \ (x, x) \in R$
Symmetric	$\forall x, y \in A \ ((x, y) \in R \rightarrow (y, x) \in R)$
Transitive	$\forall x, y, z \in A \ (((x, y) \in R \wedge (y, z) \in R) \rightarrow (x, z) \in R)$
Antisymmetric	$\forall x, y \in A \ (((x, y) \in R \wedge (y, x) \in R) \rightarrow x = y)$
Strongly connected (Total)	$\forall x, y \in A \ ((x, y) \in R \vee (y, x) \in R)$

Definition 1.4.2. A binary relation is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Definition 1.4.3. Given an equivalence relation $\sim$, we define the *equivalence class* of x under $\sim$ as: $[x]_\sim := \{y \mid x \sim y\}$.

Definition 1.4.4. Given an equivalence relation $\sim \subseteq X \times X$, we define the *quotient set* of X by $\sim$ as: $X/\sim := \{[x]_\sim \mid x \in X\}$.

Definition 1.4.5. A *partition* of a set X is a set of sets Y such that $\bigcup Y = X$, $\emptyset \notin Y$, and $\forall y_1, y_2 \in Y$ ($y_1 \neq y_2 \rightarrow y_1 \cap y_2 = \emptyset$).

Example 1.4.6. Let $\equiv_2 := \{(x, y) \in \mathbb{Z}^2 \mid x - y \text{ is even}\}$. Then $[0]_{\equiv_2} = [2]_{\equiv_2} = \{0, 2, -2, 4, -4, \dots\}$ and $[1]_{\equiv_2} = [3]_{\equiv_2} = \{1, -1, 3, -3, \dots\}$. Thus, $\mathbb{Z}/\equiv_2 = \{[0]_{\equiv_2}, [1]_{\equiv_2}\}$. Notice that this set is a partition of $\mathbb{Z}$.

Exercise 1.35. Prove that any equivalence relation on X induces a partition on X , and every partition on X induces a unique equivalence relation on X .

Commentary. Exercise 1.35 demonstrates that any equivalence relation is equivalent to specifying which elements are identified. It might seem hard to grasp at first, but this is how we naturally abstract information: we see red meat, red wine, white arms, and we, after getting enough input, group in our heads all objects that are of the same colour and associate this collection with the colour. This carries the same information as taking a quotient: we first group objects by some property that is common to them and then ignore their individual differences. Every element of one group is then identified with the others.

1.5 Basic Order Theory

Definition 1.5.1. With any binary relation $R \subseteq A \times A$, the *strict relation* corresponding to R is $R \setminus \Delta_A$, and the *weak relation* corresponding to R is $R \cup \Delta_A$.

Exercise 1.36. Let S and T be the weak and strict relations corresponding to R respectively. Prove that $\forall x, y \in A$ ($xSy \iff xTy \vee x = y$). Explain why it also follows that $\forall x, y \in A$ ($xTy \iff xSy \wedge x \neq y$).

Definition 1.5.2. A *partially ordered set* (or *poset*) is a pair $(X, \leq)$ where $\leq \subseteq X \times X$ is a binary relation that is reflexive, antisymmetric, and transitive. We call $\leq$ a (weak) *partial order*, and we denote the strict relation corresponding to it by $<$.

Definition 1.5.3. A partial order $\leq$ is a *total order* (also *linear order*) if $\leq$ is strongly connected. We call the partially ordered set with $\leq$ a total order a *totally ordered set* (or *toset*).

Exercise 1.37. Given a partial order $\leq$, explain why there is only one partially ordered set $(X, \preceq)$ such that $\preceq = \leq$.

Remark 1.5.4. Exercise 1.37 shows that explicitly defining a partially ordered set as a pair is technically redundant, as the partial order relation $\leq$ implicitly carries all the necessary information about the underlying set.

Exercise 1.38. Given a set A , prove that $(A, \{(x, y) \in A^2 \mid x \subseteq y\})$ is a partial order.

Exercise 1.39. Prove that $(\mathbb{N}, \{(x, y) \in \mathbb{N}^2 \mid \exists z \in \mathbb{N} \ z + x = y\})$ is a total order.

Definition 1.5.5. Let $(X, \leq)$ be a partially ordered set and $a \in X$. Then $X_a := \{x \in X \mid x < a\}$ is called the (proper) *initial segment* determined by a . We also define X to be

its own initial segment (but not proper) and $(X, \leq)_a := (X_a, \leq|_{X_a})$.

Definition 1.5.6. Let $(X, \leq)$ be a partially ordered set. An element $x \in X$ is said to be a:

Property	Condition
Greatest element of X	$\forall y \in X \ y \leq x$
Least element of X	$\forall y \in X \ x \leq y$
Maximal element of X	$\forall y \in X \ (x \leq y \rightarrow x = y)$
Minimal element of X	$\forall y \in X \ (y \leq x \rightarrow x = y)$
Property	Condition for a subset $S \subseteq X$
Upper bound of S	$\forall y \in S \ y \leq x$
Lower bound of S	$\forall y \in S \ x \leq y$
Supremum of S	x is the least upper bound of S
Infimum of S	x is the greatest lower bound of S

Definition 1.5.7. Let $(X, \leq)$ be a partially ordered set and $S \subseteq X$. If $(S, \leq|_S)$ has a least element x , we define $\min_{\leq} S := x$. If it has a greatest element y , we define $\max_{\leq} S := y$.

Exercise 1.40. Let $(X, \leq)$ be a partially ordered set. Prove that the least and greatest elements of a subset, if they exist, are unique. This shows that Definition 1.5.7 is mathematically unambiguous.

Definition 1.5.8. Let $(X, \leq)$ be a partially ordered set. A subset $\mathcal{C} \subseteq X$ is a chain if for any $x, y \in \mathcal{C}$ $x < y \vee x = y \vee x > y$. For brevity, we also define $\text{Chains}(X) := \{\mathcal{C} \subseteq X \mid \mathcal{C} \text{ is a chain}\}$.

Definition 1.5.9. Let $(X, \leq)$ be a partially ordered set. We say $\leq$ is a *well-ordering* if every nonempty subset of X contains a least element. We then call $(X, \leq)$ a *well-ordered set* (or *woset*).

Exercise 1.41. Prove that every well-ordered set is totally ordered.

Definition 1.5.10. Let $(X, \leq)$ be a well-ordered set. If the set $\{y \in X \mid x < y\}$ is nonempty, we define the *successor* $\text{Succ}_{\leq} x$ to be its least element.

Definition 1.5.11. A binary relation $R \subseteq A \times A$ is *well-founded* if every nonempty subset of A has a minimal element with respect to R .

Exercise 1.42. Let $\leq$ be a well-ordering. Prove that the corresponding strict relation $<$ is well-founded.

Definition 1.5.12. Let $(X, \leq)$ be a partially ordered set and $A \subseteq X$. A is (in X):

- *bounded below* if $\exists x \in X \ \forall a \in A \ x \leq a$
- *bounded above* if $\exists x \in X \ \forall a \in A \ a \leq x$

- *bounded* if it is bounded below and above in X
- *unbounded* if it is not bounded in X
- *cofinal* if $\forall x \in X \exists a \in A x \leq a$
- *dense* if $\forall x < y \in X \exists a \in A x < a < y$

1.6 Basic Lattice Theory

Definition 1.6.1. A *lattice* is a triple $(L, \wedge, \vee)$, where L is a set and $\wedge, \vee: L \times L \longrightarrow L$ are binary operations (*meet* and *join*) such that for all $x, y, z \in L$:

- (L1) **Commutativity:** $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$.
 (L2) **Associativity:** $(x \wedge y) \wedge z = x \wedge (y \wedge z)$ and $(x \vee y) \vee z = x \vee (y \vee z)$.
 (L3) **Absorption:** $x \wedge (x \vee y) = x$ and $x \vee (x \wedge y) = x$.

Exercise 1.43. Let $(L, \wedge, \vee)$ be a lattice. Prove that for every $x \in L$, $x \wedge x = x$ and $x \vee x = x$.

Definition 1.6.2. Let $(L, \wedge, \vee)$ be a lattice. We define a binary relation $\leq$ on L by: $x \leq y := x \wedge y = x$.

Exercise 1.44. Prove that $x \leq y \iff x \vee y = y$.

Theorem 1.6.3. For a lattice $(L, \wedge, \vee)$, $\leq$ is a partial order on L , and for all $x, y \in L$: $x \wedge y = \inf\{x, y\}$ and $x \vee y = \sup\{x, y\}$ (with respect to $\leq$).

Exercise 1.45. Prove Theorem 1.6.3.

Remark 1.6.4. Theorem 1.6.3 shows that a lattice, is exactly a partially ordered set in which every pair of elements has an infimum and a supremum, with $\wedge$ and $\vee$ computing them. From now on we freely use order-theoretic language for a lattice L ($\leq$, upper/lower bound, sup, inf), always meaning the order of Theorem 1.6.3.

Definition 1.6.5. A lattice L is *complete* if every subset $S \subseteq L$ has a supremum $\sup S$ and an infimum $\inf S$ in L .

Exercise 1.46. Prove that every complete lattice has a least element $0 := \inf L = \sup \emptyset$ and a greatest element $1 := \sup L = \inf \emptyset$.

Remark 1.6.6. It follows from Exercise 1.46 that a complete lattice $(L, \wedge, \vee)$ must have a nonempty L .

Exercise 1.47. Prove that for any set X , $(\mathcal{P}(X), \cap, \cup)$ is a complete lattice. What are $\sup S$, $\inf S$, and $\leq$?

Definition 1.6.7. Let L be a lattice and $f: L \longrightarrow L$. We say f is *monotone* if $\forall x, y \in L (x \leq y \rightarrow f(x) \leq f(y))$. A *fixed point* of f is an element $x \in L$ with $f(x) = x$.

Theorem 1.6.8 (Knaster–Tarski). *Let $(L, \wedge, \vee)$ be a complete lattice and $f: L \rightarrow L$ be monotone. Then f has a least fixed point.*

Proof. Let $A := \{x \in L \mid f(x) \leq x\}$ and $a := \inf A$. Pick any $x \in A$, then $a \leq x$, so $f(a) \leq f(x) \leq x$ and thus $f(a) \leq \inf A = a$, showing that $a \in A$ and $f(a) \in A$. Since $f(a) \in A$, we get that $a \leq f(a)$, so $f(a) = a$. Let b be a fixed point of f , then $f(b) \leq b$, so $b \in A$ and thus $a \leq b$. **Q.E.D.**

Definition 1.6.9. Let $(L, \wedge, \vee)$ and f be as in Theorem 1.6.8. We denote the least fixed point of f as μf .

Exercise 1.48. Prove that if $(L, \leq)$ is a complete lattice, then so is $(L, \geq)$ (the order-dual), with sup and inf exchanged.

Exercise 1.49. Deduce from the previous exercise and Theorem 1.6.8 that f also has a *greatest* fixed point, namely $\nu f := \sup\{x \in L \mid x \leq f(x)\}$.

Theorem 1.6.10 (General Induction Principle). *Let X be a set and $F: \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ be monotone. Let $S^* := \mu F$ be its least fixed point, guaranteed by Theorem 1.6.8. Then for any $T \subseteq X$, $F(T) \subseteq T \implies S^* \subseteq T$.*

Proof. By Theorem 1.6.8 and the example above, $S^* = \inf\{S \subseteq X \mid F(S) \subseteq S\} = \bigcap\{S \subseteq X \mid F(S) \subseteq S\}$. If $F(T) \subseteq T$, then T is one of the sets in this intersection, so $S^* \subseteq T$. **Q.E.D.**

Commentary. Theorem 1.6.10 is the foundational engine behind every induction principle in mathematics. Whenever we define a collection as the “smallest” set satisfying certain rules, we dream of an ideal form. All too human — recall Pygmalion who, dissatisfied with the imperfections of the world, carved a figure out of ivory so pristine that the gods were moved to breathe life into it.

We do not conjure such minimal sets out of thin air. Instead, the theorem provides the sculptor’s chisel: we start with the entire ambient universe as a rough block of stone and intersect every closed set, chipping away every extraneous element until only the minimal, essential core S^* remains.

Chapter 2

Formal Languages

2.1 Formal Languages

Before analyzing the syntax of propositional or first-order logic, we construct a general theory of formal languages and strings.

Definition 2.1.1. An *alphabet* Σ is a set of symbols, also called letters.

Definition 2.1.2. We define the set of *strings* over Σ , denoted Σ^* , as the smallest set satisfying:

1. The empty string, ε , is in Σ^* .
2. If $\sigma \in \Sigma^*$ and $s \in \Sigma$, the string σs obtained by appending s to σ is in Σ^* .

Theorem 2.1.3 (Induction on strings). *Let Π be a property of strings over Σ that satisfies:*

1. Π holds for ε .
2. If Π holds for σ , then for any $s \in \Sigma$, Π holds for σs .

Then Π holds for all strings over Σ .

Proof. Let S be the set of strings over Σ for which Π holds. Then S satisfies conditions 1 and 2 in Definition 2.1.2. Since Σ^* is the smallest set satisfying these conditions, we must have $\Sigma^* \subseteq S$. ***Q.E.D.***

Definition 2.1.4. Given two strings σ, τ over Σ , we define their *concatenation* $\sigma\tau$ recursively as follows:

1. If $\tau = \varepsilon$, then $\sigma\tau := \sigma$.
2. If $\tau = \tau't$ for some $t \in \Sigma$, then $\sigma\tau := (\sigma\tau')t$.

Exercise 2.1. Prove that $|\sigma\tau| = |\sigma| + |\tau|$, where $|\sigma|$ denotes the length of the string σ .

Definition 2.1.5. A *language* $\mathcal{L}$ over Σ is any subset of Σ^* .

Example 2.1.6. Let $\Sigma := \{0, 1, \dots, 9\}$ and let $\mathcal{L}$ be defined recursively as follows: $0, 1, \dots, 9$ are in $\mathcal{L}$. If $S \in \mathcal{L}$, then $1S, \dots, 9S$ are in $\mathcal{L}$.

Exercise 2.2. Describe the language $\mathcal{L}$ in the example above in one simple sentence, without mentioning the recursive rules.

Example 2.1.7. Let $\Sigma := \{ (,) \}$ and let $\mathcal{L}$ be defined recursively by: $()$ is in $\mathcal{L}$. If $S \in \mathcal{L}$, then $()S$, (S) , and $S()$ are in $\mathcal{L}$.

Exercise 2.3. Describe the language $\mathcal{L}$ in the example above in one sentence.

Example 2.1.8. Let Σ consist of all letters of the English alphabet, and let $\mathcal{L}$ be defined recursively as follows: *man, woman, dog, cat* are nouns. *big, small* are adjectives. *walks, runs, eats, drinks* are verbs. If A is an adjective, N is a noun, and V is a verb, then $A N V$ and $N V$ are sentences. If S and T are sentences, then $S \text{ and } T$ and $S \text{ or } T$ are sentences.

Exercise 2.4. Write down three valid sentences belonging to the language $\mathcal{L}$ defined in the example above.

Definition 2.1.9. Let σ and π be strings over Σ . We say σ is an *initial segment* of π if there exists a string $\tau \in \Sigma^*$ such that $\pi = \sigma\tau$.

Exercise 2.5. Prove that the empty string ε is an initial segment of every string.

Exercise 2.6. Let σ be an initial segment of τ , and let τ be an initial segment of σ . Prove that $\sigma = \tau$.

Chapter 3

Propositional Calculus

3.1 Syntax of Propositional Calculus

We now apply our string and language definitions to define the syntax of Propositional Calculus.

Definition 3.1.1. Let $\mathbf{Var} := \{p_1, p_2, \dots\}$ be a set of variables, and let $\mathbf{Con} := \{c_1^0, c_2^0, \dots, c_1^1, \dots\}$ be a set of logical connectives, where the superscript denotes the arity. The set of formulae $\mathbf{Frm}$ is defined as the smallest set satisfying:

1. All variables in $\mathbf{Var}$ are in $\mathbf{Frm}$.
2. If $\varphi_1, \dots, \varphi_n \in \mathbf{Frm}$ and c_k^n is an n -ary connective, then the string $c_k^n(\varphi_1, \dots, \varphi_n)$ is in $\mathbf{Frm}$.

Remark 3.1.2. For finite sets, “smallest” simply means having the minimum cardinality. However, when working with infinite sets of formulae, defining sets by counting their elements is often futile.

Exercise 3.1. Prove that every formula $\varphi \in \mathbf{Frm}$ can be obtained by a finite number of applications of the syntactic rules.

Example 3.1.3. 1. p_1 is a formula.

2. $c_3^2(p_1, p_4)$ is a formula.

3. $c_1^3(c_2^0, p_1, c_1^3(p_3, p_2, c_4^1(p_2)))$ is a formula.

Definition 3.1.4. The *complexity* of a propositional formula is defined recursively as follows:

1. The complexity of any variable $p \in \mathbf{Var}$ and any 0-ary connective c_n^0 is 0.
2. The complexity of $c_m^n(\varphi_1, \dots, \varphi_n)$ is defined as $\max\{\text{comp}(\varphi_1), \dots, \text{comp}(\varphi_n)\} + 1$.

Example 3.1.5. Since the complexity of q and r is 0, the complexity of $q \wedge r$ is 1. Since the complexity of p is 0, the complexity of $p \rightarrow (q \wedge r)$ is 2.

Remark 3.1.6. Notice that the complexity of any proper subformula is strictly less than the complexity of the parent formula.

Theorem 3.1.7 (Induction on propositional formulae). *Let Π be a property of propositional formulae such that:*

1. Π holds for all variables.
2. If Π holds for $\varphi_1, \dots, \varphi_n$, then Π holds for $c_m^n(\varphi_1, \dots, \varphi_n)$ for any n, m .

*Then Π holds for all propositional formulae in **Frm**.*

Proof. Let Π be given as in the theorem. Let $\mathcal{C} \subseteq \mathbf{Frm}$ be the set of formulae for which Π holds. Condition 1 gives us $\mathbf{Var} \subseteq \mathcal{C}$ and if $\varphi_1, \dots, \varphi_n \in \mathcal{C}$ and c_k^n is an n -ary connective, then condition 2 gives us $c_k^n(\varphi_1, \dots, \varphi_n) \in \mathcal{C}$. Thus $\mathcal{C}$ satisfies the conditions defining **Frm** and because **Frm** is the smallest set satisfying these conditions, we conclude $\mathbf{Frm} \subseteq \mathcal{C}$. **Q.E.D.**

Intuition. The validity of this induction principle is intuitive when we trace a formula's construction: consider $c_3^2(x, c_1^1(y))$. By the base case, Π holds for the variables x and y . Applying the inductive step, Π must hold for $c_1^1(y)$, and applying it once more, it must hold for $c_3^2(x, c_1^1(y))$.

Let us see a concrete example of how to use this induction principle to prove a syntactic property.

Example 3.1.8 (Finiteness of Variable Occurrences). Let $V(\varphi)$ denote the set of variables that occur in a formula φ . We want to prove that $V(\varphi)$ is a finite set for every formula $\varphi \in \mathbf{Frm}$.

Proof. We proceed by induction on the structure of propositional formulae.

Base Case: Let φ be a variable p . Then $V(p) = \{p\}$. Since a singleton set is finite, the property holds.

Inductive Step: Let $\varphi_1, \dots, \varphi_n$ be formulae, and assume the inductive hypothesis: $V(\varphi_1), \dots, V(\varphi_n)$ are all finite sets. Let c_k^n be an n -ary connective. We evaluate the formula $\psi = c_k^n(\varphi_1, \dots, \varphi_n)$. By definition, the variables in ψ are exactly those that occur in any of its subformulae:

$$V(\psi) = V(\varphi_1) \cup \dots \cup V(\varphi_n)$$

Since the union of finitely many finite sets is finite, $V(\psi)$ is finite. By the induction principle, $V(\varphi)$ is finite for every formula φ . **Q.E.D.**

Exercise 3.2. Prove that no formula contains more variables than the total number of symbols in its string representation.

Exercise 3.3. A foo-linked list is defined recursively as follows:

1. $[]$ is a foo-linked list.
2. If L is a foo-linked list and x is of type foo, then $x :: L$ is a foo-linked list.

The set of all foo-linked lists is the smallest set satisfying these rules. State an induction rule for foo-linked lists analogous to the induction rule for propositional formulae.

Exercise 3.4. Prove your stated induction rule for foo-linked lists.

Exercise 3.5. ($\star$) The concatenation of two foo-linked lists is defined recursively by:

1. $[]@L := L$.
2. $(x :: L')@L := x :: (L'@L)$.

The length of a foo-linked list is defined recursively by:

1. $l([]) := 0$.
2. $l(x :: L) := 1 + l(L)$.

Prove that $l(L'@L) = l(L') + l(L)$ (you may assume basic properties of arithmetic).

3.2 Semantics of Propositional Calculus

Now that set theory is established, we can define interpretations formally as functions rather than informal “assignments”.

Definition 3.2.1. An *interpretation* (or *valuation*) is a function $\mathcal{I}: \mathbf{Var} \rightarrow \{\text{False}, \text{True}\}$.

Definition 3.2.2. An *extended interpretation* corresponding to an interpretation $\mathcal{I}$ is a function $\hat{\mathcal{I}}: \mathbf{Frm} \rightarrow \{\text{False}, \text{True}\}$ that agrees with $\mathcal{I}$ on all variables.

Definition 3.2.3. Given an interpretation $\mathcal{I}$, the *standard extended interpretation* $\hat{\mathcal{I}}$ evaluates formulae according to the following truth tables:

$\hat{\mathcal{I}}(\perp)$ $\hat{\mathcal{I}}(\top)$		$\hat{\mathcal{I}}(\phi)$ $\hat{\mathcal{I}}(\neg\phi)$			
False	True	False	True	True	False
$\hat{\mathcal{I}}(\phi)$	$\hat{\mathcal{I}}(\psi)$	$\hat{\mathcal{I}}(\phi \wedge \psi)$	$\hat{\mathcal{I}}(\phi \vee \psi)$	$\hat{\mathcal{I}}(\phi \rightarrow \psi)$	$\hat{\mathcal{I}}(\phi \iff \psi)$
False	False	False	False	True	True
False	True	False	True	True	False
True	False	False	True	False	False
True	True	True	True	True	True

Remark 3.2.4 (The Intuition of Material Implication). The truth table for the conditional $\phi \rightarrow \psi$ (material implication) is often counterintuitive to beginners, particularly the rows where the premise ϕ is false. In classical logic, “if ϕ then ψ ” is treated as a contract or a promise. The only way to *break* this promise is if ϕ occurs, but ψ fails to happen (the third row of the table). If ϕ is false, the condition of the promise was never met; therefore, the promise has not been broken, and the statement is classified as *vacuously true*. For example, the statement “If it rains, I will open my umbrella” remains true on a sunny day, regardless of whether my umbrella is open or closed.

Remark 3.2.5. Strictly speaking, the interpretation $\mathcal{I}$ and its extended counterpart $\hat{\mathcal{I}}$ are distinct functions. However, because logicians are primarily interested in $\hat{\mathcal{I}}$, we conventionally omit the circumflex and refer to both as simply the “interpretation” $\mathcal{I}$. Since they agree on the intersection of their domains, this causes no formal issues.

Definition 3.2.6. Two formulae φ and ψ are *equivalent*, written $\varphi \equiv \psi$, if $\mathcal{I}(\varphi) = \mathcal{I}(\psi)$ for all interpretations $\mathcal{I}$.

Exercise 3.6. Prove that two distinct variables are never equivalent.

Definition 3.2.7. A formula φ is a *tautology* if $\varphi \equiv \top$.

Definition 3.2.8. A formula φ is a *contradiction* if $\varphi \equiv \perp$.

Definition 3.2.9. A formula φ is *satisfiable* if there exists some interpretation $\mathcal{I}$ such that $\mathcal{I}(\varphi) = \text{True}$.

Definition 3.2.10. An interpretation $\mathcal{I}$ *satisfies* a set of formulae Σ if it satisfies every formula in Σ .

Definition 3.2.11. A set of formulae Σ is *satisfiable* if there exists some interpretation $\mathcal{I}$ such that $\mathcal{I}(\varphi) = \text{True}$ for all $\varphi \in \Sigma$.

Definition 3.2.12. A set of formulae Σ *entails* φ , written $\Sigma \models \varphi$, if every interpretation that satisfies Σ also satisfies φ .

Definition 3.2.13. $\phi \models \psi$ if $\{\phi\} \models \psi$.

Definition 3.2.14. $\models \phi$ if $\emptyset \models \phi$.

Exercise 3.7. Prove that $\varphi \equiv \varphi$ for any formula φ .

Exercise 3.8. Prove that $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

Exercise 3.9. Explain why, given that ψ is false, the reasoning: “If φ then ψ , but not ψ , therefore not φ ” is correct.

Exercise 3.10. Prove that $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

Exercise 3.11. Prove that $\neg(p \rightarrow q) \equiv p \wedge \neg q$.

Exercise 3.12. Prove that $\phi \models \psi$ and $\psi \models \phi$ if and only if $\phi \equiv \psi$.

Exercise 3.13. Prove that if $\Sigma \models \phi \rightarrow \psi$ and $\Sigma \models \phi$, then $\Sigma \models \psi$.

Exercise 3.14. Prove that if $\phi \in \Sigma$, then $\Sigma \models \phi$.

Exercise 3.15. Prove that ϕ is a tautology if and only if $\models \phi$.

Exercise 3.16. Prove that φ is satisfiable if and only if it is not a contradiction.

Exercise 3.17. Given a formula φ containing n distinct variables, how many distinct interpretations exist over these variables?

Definition 3.2.15. A set of connectives $\mathcal{C}$ is *functionally complete* if every possible formula is equivalent to some formula containing only connectives from $\mathcal{C}$.

Example 3.2.16. The set $\{\neg, \wedge, \vee\}$ is functionally complete. For instance: $\top \equiv p \vee (\neg p)$. $p \rightarrow q \equiv \neg p \vee q$. A ternary connective c_0^3 that evaluates to true under $\mathcal{I}$ if and only if p is true, or if p is false but q is true, can be written as $p \vee (\neg p \wedge q)$.

Exercise 3.18. Explain why $\{\neg, \wedge, \vee\}$ can express any arbitrary n -ary connective c_m^n . *Hint:* Consider the structure of the truth table of c_m^n .

Exercise 3.19. Prove that $\{\neg, \rightarrow\}$ is functionally complete.

Exercise 3.20. $(\star\star)$ Prove that $\{\wedge, \vee, \iff\}$ is not functionally complete. *Hint:* Identify a semantic property shared by all formulae built using only these three connectives, which is not shared by every formula.

Exercise 3.21. $(\star\star)$ Find all binary connectives c_k^2 such that the singleton set $\{c_k^2\}$ is functionally complete, and prove that these are the only ones. *Hint:* Refer to Example 3.2.16.

Chapter 4

Proof Theory

4.1 Formal Proof Systems

Having defined formulae and truth, we now study deductions from a purely syntactic perspective.

Definition 4.1.1. A *proof system* is a triple $(\mathcal{L}, \mathcal{R}, \mathcal{A})$, where:

1. $\mathcal{L}$ is a formal language.
2. $\mathcal{R}$ is a set of rules of inference.
3. $\mathcal{A}$ is a set of axioms.

Definition 4.1.2. A *proof* in a proof system $(\mathcal{L}, \mathcal{R}, \mathcal{A})$ is a finite sequence of formulae in $\mathcal{L}$, where each formula is either an axiom, an assumption, or follows from preceding formulae via an inference rule.

Commentary. The definition of a proof is a formalisation of a valid logical argument that is verifiable by a (in the words of Alan Turing) “human computer” — a bookkeeper capable of only following the simplest instructions. As such, the proof must be finite, since he must be able to finish his task in a finite amount of time. The proof must consist of the simplest steps, referring only to axioms, assumptions, and previously established results.

Exercise 4.1. Let P be a proof of length n . Let Q be the sequence obtained by restricting P to its first m elements, where $m < n$. Show that Q is also a valid proof.

Remark 4.1.3. It is crucial to distinguish between mathematics and metamathematics — what we have informally called a “proof” up to this point is not the same object as the formal proofs of proof theory.

Definition 4.1.4. We write $\Sigma \vdash \phi$, read as “ Σ proves ϕ ”, if there exists a formal proof of ϕ where all active assumptions belong to Σ .

Definition 4.1.5. A formula φ is a *theorem* of a proof system if $\emptyset \vdash \varphi$, which we write simply as $\vdash \varphi$.

Exercise 4.2. Prove that if $\phi \in \Sigma$, then $\Sigma \vdash \phi$.

4.2 Axiomatic Propositional Calculus

We now explore axiomatic systems, starting with Modus Ponens and the Tautology system.

Definition 4.2.1. *Modus Ponens* is the rule of inference: From ϕ and $\phi \rightarrow \psi$, we may infer ψ .

Definition 4.2.2. The *tautology system* is a propositional proof system where $\mathcal{R}$ contains only Modus Ponens, and $\mathcal{A}$ consists of all semantic tautologies.

Example 4.2.3. We can show that $\{\phi, \psi, (\phi \wedge \psi) \rightarrow \xi\} \vdash \xi$ in this system:

Proof. 1. $\phi \rightarrow \psi \rightarrow (\phi \wedge \psi)$ (Axiom: Tautology)

2. ϕ (Assumption)

3. $\psi \rightarrow (\phi \wedge \psi)$ (Modus Ponens on 1, 2)

4. ψ (Assumption)

5. $\phi \wedge \psi$ (Modus Ponens on 3, 4)

6. $(\phi \wedge \psi) \rightarrow \xi$ (Assumption)

7. ξ (Modus Ponens on 5, 6)

Q.E.D.

Example 4.2.4 (Ex Falso Quodlibet). We show that $\{\phi, \neg\phi\} \vdash \psi$:

Proof. 1. $\phi \rightarrow \neg\phi \rightarrow \psi$ (Axiom: Tautology)

2. ϕ (Assumption)

3. $\neg\phi \rightarrow \psi$ (Modus Ponens on 1, 2)

4. $\neg\phi$ (Assumption)

5. ψ (Modus Ponens on 3, 4)

Q.E.D.

Exercise 4.3. Prove that $\neg\neg\phi \vdash \phi$ in the tautology system.

Definition 4.2.5. The *Hilbert system* is a proof system where $\mathcal{R}$ consists solely of Modus Ponens, and the axioms $\mathcal{A}$ are all formulae of the following forms:

A1. $\phi \rightarrow (\psi \rightarrow \phi)$

A2. $(\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))$

A3. $(\neg\phi \rightarrow \neg\psi) \rightarrow (\psi \rightarrow \phi)$

Definition 4.2.6. A set of formulae Γ is *inconsistent* if $\Gamma \vdash \perp$.

Definition 4.2.7. A set of formulae Γ is *consistent* if it is not inconsistent.

Definition 4.2.8. A proof system is *sound* if $\Sigma \vdash \phi \implies \Sigma \models \phi$.

Intuition. Soundness ensures that our system only proves logically true facts.

Definition 4.2.9. A proof system is *complete* if $\Sigma \models \phi \implies \Sigma \vdash \phi$.

Intuition. Completeness ensures that our system is powerful enough to prove every logical truth.

4.3 Natural Deduction for Propositional Logic

Rather than writing proofs as linear sequences of formulae starting from complex axioms, we can represent proofs as trees using Natural Deduction, which relies on introducing and eliminating connectives.

Definition 4.3.1 (Proof Tree). Let $\mathcal{L}$ be a formal language and $\mathcal{R}$ a set of inference rules. The set of **proof trees** over $\mathcal{L}$ is defined as the smallest set satisfying:

1. For any formula $\phi \in \mathcal{L}$, the tree consisting of the single node ϕ is a proof tree. The formula ϕ is called an **open assumption** of this tree.
2. If $\Pi_1, \dots, \Pi_k$ are proof trees with roots (conclusions) $\phi_1, \dots, \phi_k$, and

$$\frac{\phi_1 \quad \dots \quad \phi_k}{\psi}$$

is an inference rule in $\mathcal{R}$, then the tree obtained by placing ψ as a new root below these subtrees is a proof tree of ψ .

Definition 4.3.2 (Discharging Assumptions). Certain inference rules in Natural Deduction permit us to **discharge** (or close) open assumptions. When such a rule is applied, we enclose all leaves of the tree labeled with the discharged formula in square brackets and index them with a label n , written $[\phi]^n$. These assumptions are then **closed**.

Definition 4.3.3 (Natural Deduction Derivation). We say Σ proves ϕ in Natural Deduction, written $\Sigma \vdash \phi$, if there exists a finite proof tree with root ϕ such that every undischarged (open) assumption of the tree is an element of Σ .

The inference rules for propositional Natural Deduction are as follows:

Symbol	Introduction Rules	Elimination Rules
$\wedge$	$\frac{\phi \quad \psi}{\phi \wedge \psi} \wedge I$	$\frac{\phi \wedge \psi}{\phi} \wedge E_1 \quad \frac{\phi \wedge \psi}{\psi} \wedge E_2$
$\vee$	$\frac{\phi}{\phi \vee \psi} \vee I_1 \quad \frac{\psi}{\phi \vee \psi} \vee I_2$	$\frac{\phi \vee \psi \quad \begin{array}{c} [\phi]^n \\ \vdots \\ \chi \end{array} \quad \begin{array}{c} [\psi]^m \\ \vdots \\ \chi \end{array}}{\chi} \vee E_{n,m}$
$\rightarrow$	$\frac{\begin{array}{c} [\phi]^n \\ \vdots \\ \psi \\ \hline \phi \rightarrow \psi \\ [\phi]^n \end{array}}{\phi \rightarrow \psi} \rightarrow I_n$	$\frac{\phi \rightarrow \psi \quad \phi}{\psi} \rightarrow E$
$\neg$	$\frac{\begin{array}{c} \vdots \\ \perp \\ \hline \neg \phi \end{array}}{\neg \phi} \neg I_n$	$\frac{\neg \phi \quad \phi}{\perp} \neg E$
$\perp$	(None)	$\frac{\perp}{\phi} \perp E \quad \frac{\begin{array}{c} [\neg \phi]^n \\ \vdots \\ \perp \end{array}}{\neg \phi} \text{RAA}_n$

To see how this abstract tree-like structure works in practice, let us construct a simple Natural Deduction derivation.

Example 4.3.4 (Commutativity of Conjunction). We want to prove that $\{p \wedge q\} \vdash q \wedge p$. We build the proof tree by starting with the assumption, breaking it apart, and rebuilding it in the opposite order:

$$\frac{\frac{\frac{[p \wedge q]^1}{q} \wedge E_2 \quad \frac{[p \wedge q]^1}{p} \wedge E_1}{q \wedge p} \wedge I}{(p \wedge q) \rightarrow (q \wedge p)} \rightarrow I_1$$

Notice how the open assumption $p \wedge q$ is temporarily assumed, used twice, and then closed (discharged) at the very end when we apply the implication introduction rule $\rightarrow I_1$. The resulting theorem has zero open assumptions, proving $\vdash (p \wedge q) \rightarrow (q \wedge p)$.

Theorem 4.3.5 (Deduction Theorem). $\Gamma \cup \{\phi\} \vdash \psi \iff \Gamma \vdash \phi \rightarrow \psi$.

Proof. ($\implies$) Let Π be a proof tree of ψ from $\Gamma \cup \{\phi\}$. By applying the $\rightarrow I$ rule, we obtain a proof tree of $\phi \rightarrow \psi$ where the assumption ϕ is discharged. The remaining open assumptions are in Γ .

($\impliedby$) Let Π be a proof tree of $\phi \rightarrow \psi$ from Γ . By extending this tree with an open assumption ϕ and applying $\rightarrow E$, we obtain a proof tree of ψ with open assumptions in $\Gamma \cup \{\phi\}$. **Q.E.D.**

Chapter 5

First-Order Logic

5.1 What First-Order Logic Should Be

Propositional Calculus is not enough to encompass even basic reasoning. For example, if we represent “Socrates is a man” by P and “Socrates is mortal” by Q , we cannot represent the connection between “man” and “mortal”. First-Order Logic solves this by looking inside sentences, introducing *objects* (like Socrates), *properties/relations* (like being a man or being mortal), and *quantifiers* (which tell us how many objects satisfy a property).

5.1.1 Quantifiers

The two First-Order quantifiers are:

- **The Universal Quantifier ($\forall$):** Read as “for all”, “every”, or “(for) each”. The statement $\forall x P(x)$ asserts that *every* object under consideration has the property P .
- **The Existential Quantifier ($\exists$):** Read as “there exists”, “(there exist) some”, or “at least one”. The statement $\exists x P(x)$ asserts that there is *at least one* object under consideration that has the property P .

Crucially, quantifiers always range over a set of objects called the **universe of discourse** (or **domain**).

Example 5.1.1 (Semantics depends on the universe). The sentence $\forall x x \geq 0$ is true in $\mathbb{N}$, but false in $\mathbb{Z}$.

5.2 First-Order Syntax

We now apply our intuitive definitions to build the syntax of First-Order Logic formally.

Definition 5.2.1. A *signature* σ is a quadruple $(\mathbf{F}, \mathbf{R}, \mathbf{C}, \text{ar})$, where $\mathbf{F}$, $\mathbf{R}$, and $\mathbf{C}$ are pairwise disjoint sets, and no element of any of them is an initial segment of another. The function ar assigns a natural number (representing arity) to every element of $\mathbf{F}$ and $\mathbf{R}$.

Definition 5.2.2. The alphabet of First-Order Logic over a signature σ consists of:

- **Logical symbols:**

1. Connectives: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$
2. Quantifiers: $\forall, \exists$
- **Variables:** $x_0, x_1, \dots$
- **Non-logical symbols:**
 1. Function symbols: Elements of $\mathbf{F}$.
 2. Relation symbols: Elements of $\mathbf{R}$.
 3. Constant symbols: Elements of $\mathbf{C}$.

Definition 5.2.3. The set of *terms* $\mathbf{Trm}(\sigma)$ over σ is the smallest set satisfying:

1. All constant symbols and variables are in $\mathbf{Trm}(\sigma)$.
2. If $t_1, \dots, t_n \in \mathbf{Trm}(\sigma)$ and $f \in \mathbf{F}$ is an n -ary function symbol, then the string $ft_1 \dots t_n$ is in $\mathbf{Trm}(\sigma)$.

Theorem 5.2.4 (Induction for terms). *Let Π be a property of terms over σ satisfying:*

1. Π holds for all constant symbols and variables.
2. If Π holds for $t_1, \dots, t_n$ and f is an n -ary function symbol, then Π holds for $ft_1 \dots t_n$.

Then Π holds for all terms in $\mathbf{Trm}(\sigma)$.

Exercise 5.1. Prove Theorem 5.2.4.

Definition 5.2.5. For a term $t \in \mathbf{Trm}(\sigma)$, we define $\mathbf{Var}(t)$ to be the set of variables used in t .

Definition 5.2.6. The sets $\mathbf{AtFrm}(\sigma)$ (atomic formulae) and $\mathbf{Frm}(\sigma)$ (formulae) over σ are the smallest sets satisfying:

1. **Atomic formulae:**
 - (a) If $t_1, \dots, t_n \in \mathbf{Trm}(\sigma)$ and $R \in \mathbf{R}$ is an n -ary relation symbol, then $Rt_1 \dots t_n \in \mathbf{AtFrm}(\sigma)$.
 - (b) If $t_1, t_2 \in \mathbf{Trm}(\sigma)$, then $t_1 t_2 \in \mathbf{AtFrm}(\sigma)$.
2. **Inclusion:** $\mathbf{AtFrm}(\sigma) \subseteq \mathbf{Frm}(\sigma)$.
3. **Compound formulae:** If $\varphi, \psi \in \mathbf{Frm}(\sigma)$, then the following strings are in $\mathbf{Frm}(\sigma)$: $\neg\varphi, \wedge\varphi\psi, \vee\varphi\psi, \rightarrow\varphi\psi, \leftrightarrow\varphi\psi, \forall x\varphi$, and $\exists x\varphi$.

Theorem 5.2.7 (Induction for FOL formulae). *Let Π be a property of First-Order formulae over σ satisfying:*

1. Π holds for all atomic formulae.
2. If Π holds for φ and ψ , then Π holds for $\neg\varphi, \wedge\varphi\psi, \vee\varphi\psi, \rightarrow\varphi\psi, \leftrightarrow\varphi\psi, \forall x\varphi$, and $\exists x\varphi$.

Then Π holds for all formulae in $\mathbf{Frm}(\sigma)$.

Exercise 5.2. Prove Theorem 5.2.7.

To show that formulae are uniquely readable, we introduce a formal symbol weight assignment.

Definition 5.2.8. The weight assignment w for FOL symbols is defined as:

Symbol Type	Symbol(s)	Weight $w(s)$
n -ary Function Symbol	$f \in \mathbf{F}$	$\text{ar } f - 1$
n -ary Relation Symbol	$R \in \mathbf{R}$	$\text{ar } R - 1$
Constant Symbol	$c \in \mathbf{C}$	-1
Variable	$x_0, x_1, \dots$	-1
Equality Symbol	$=$	1
Unary Connective	$\neg$	0
Binary Connective	$\wedge, \vee, \rightarrow, \leftrightarrow$	1
Quantifier	$\forall, \exists$	1

We extend this to strings σ by defining: $\hat{w}(\varepsilon) := 0$, and $\hat{w}(\sigma s) := \hat{w}(\sigma) + w(s)$.

Example 5.2.9 (Calculating String Weights). To see how the extended weight assignment works, let us calculate $\hat{w}(\sigma)$ for the Polish notation formula $\sigma = \wedge p \neg q$. Let us evaluate the cumulative weight of each initial segment of the string:

$$\begin{aligned}
 \hat{w}(\varepsilon) &= 0 \\
 \hat{w}(\wedge) &= 0 + w(\wedge) = 0 + 1 = 1 \\
 \hat{w}(\wedge p) &= 1 + w(p) = 1 + (-1) = 0 \\
 \hat{w}(\wedge p \neg) &= 0 + w(\neg) = 0 + 0 = 0 \\
 \hat{w}(\wedge p \neg q) &= 0 + w(q) = 0 + (-1) = -1
 \end{aligned}$$

Notice that the final weight of the complete formula is -1 , and every proper initial segment of the string (like $\wedge$ or $\wedge p \neg$) has a weight greater than or equal to 0 . This is precisely the property generalized in Exercises 5.3 and 5.4.

Exercise 5.3. Prove that if $t \in \mathbf{Trm}(\sigma)$, then $\hat{w}(t) = -1$, and if t is a proper initial segment of a term, then $\hat{w}(t) \geq 0$.

Exercise 5.4. Prove that if $\varphi \in \mathbf{Frm}(\sigma)$, then $\hat{w}(\varphi) = -1$, and if φ is a proper initial segment of a formula, then $\hat{w}(\varphi) \geq 0$.

Remark 5.2.10. Exercises 5.3 and 5.4 show that terms and formulae written in prefix notation (Polish notation) are uniquely readable.

Commentary. We have used Polish notation (developed by Jan Łukasiewicz) to prove unique readability. Once the cathedral is built, the scaffolding is removed; in a similar manner, we return to the much uglier infix notation merely because the mathematical community at large employs it. We do not need to prove that the infix notation is uniquely readable, since every formula is actually still in Polish notation in its “soul” — we simply write it differently.

Definition 5.2.11. The set of free variables of a formula φ : $\mathbf{FV}(\varphi)$ is defined recursively by:

- $\mathbf{FV}(Rt_1 \dots t_n) := \bigcup_{1 \leq k \leq n} \mathbf{FV}(t_k)$
- $\mathbf{FV}(= t_1 t_2) := \mathbf{FV}(t_1) \cup \mathbf{FV}(t_2)$
- $\mathbf{FV}(\neg \varphi) := \mathbf{FV}(\varphi)$
- $\mathbf{FV}(*\varphi\psi) := \mathbf{FV}(\varphi) \cup \mathbf{FV}(\psi)$, where $*$ $\in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$
- $\mathbf{FV}(*x \varphi) := \mathbf{FV}(\varphi) \setminus \{x\}$, where $*$ $\in \{\exists, \forall\}$

Definition 5.2.12. We define standard abbreviation conventions, under the assumption that the variables in the tuple $\bar{x}$ are pairwise distinct and do not occur bound in the outer scope of φ :

1. $\varphi(\bar{x})$ abbreviates $\varphi(x_1, \dots, x_n)$.
2. $\forall \bar{x} \varphi$ and $\exists \bar{x} \varphi$ abbreviate iterated quantifiers.
3. $\forall R(\bar{x}) \varphi := \forall \bar{x} (R(\bar{x}) \rightarrow \varphi)$, where only the variables in $\bar{x}$ are bound by the quantifier, and any other free variables in R act as constant parameters.
4. $\exists R(\bar{x}) \varphi := \exists \bar{x} (R(\bar{x}) \wedge \varphi)$, where only the variables in $\bar{x}$ are bound, and any other free variables in R act as constant parameters.
5. $\exists! x \varphi(x) := \exists x (\varphi(x) \wedge \forall y (\varphi(y) \rightarrow x = y))$, where y is a variable that does not occur in φ .

Definition 5.2.13. To avoid writing excess parentheses, we establish operator precedence:

Precedence	Operators
1	$\neg$
2	$\wedge, \vee$
3	$\forall \bar{x}, \exists \bar{x}, \forall R(\bar{x}), \exists R(\bar{x})$
4	$\rightarrow, \leftrightarrow$

Furthermore, we declare all binary connectives to be right-associative.

Example 5.2.14. The formula $\neg \exists P(\bar{x}) R(\bar{x}) \wedge S(x) \rightarrow x = x$, where $\bar{x}$ is the sequence x, y parses as: $(\neg \exists x \exists y (P(x, y) \wedge (R(x, y) \wedge S(x)))) \rightarrow (x = x)$.

Definition 5.2.15. A formula $\varphi(\bar{x}, y)$ is *functional* (in y) if $\forall \bar{x} \exists! y \varphi(\bar{x}, y)$.

5.3 First-Order Semantics

Having established functions and relations, we can define first-order models and satisfaction rigorously.

Definition 5.3.1. A *structure* $\mathfrak{A}$ over a signature $\sigma = (\mathbf{F}, \mathbf{R}, \mathbf{C}, \text{ar})$ is a pair $(A, \cdot^{\mathfrak{A}})$, where $A \neq \emptyset$ is a set called the *universe* (often denoted $|\mathfrak{A}|$), and $\cdot^{\mathfrak{A}}$ is an interpretation function mapping:

1. Each n -ary function symbol $f \in \mathbf{F}$ to a function $f^{\mathfrak{A}}: A^n \rightarrow A$.
2. Each n -ary relation symbol $R \in \mathbf{R}$ to a relation $R^{\mathfrak{A}} \subseteq A^n$.
3. Each constant symbol $c \in \mathbf{C}$ to an element $c^{\mathfrak{A}} \in A$.

Definition 5.3.2. A *variable assignment* in $\mathfrak{A}$ is a function $s: \mathbf{Var} \rightarrow |\mathfrak{A}|$.

Definition 5.3.3. The *valuation function* $\bar{s}: \mathbf{Trm}(\sigma) \rightarrow |\mathfrak{A}|$ corresponding to an assignment s is defined recursively by:

1. $\bar{s}(x_i) := s(x_i)$.
2. $\bar{s}(c) := c^{\mathfrak{A}}$.
3. $\bar{s}(ft_1 \dots t_n) := f^{\mathfrak{A}}(\bar{s}(t_1), \dots, \bar{s}(t_n))$.

Definition 5.3.4. We define the modified assignment $s_{x \mapsto t}$ by: $s_{x \mapsto t}(y) := \begin{cases} s(y) & \text{if } y \neq x, \\ t & \text{if } y = x. \end{cases}$

Definition 5.3.5. The *satisfaction* relation $\mathfrak{A} \models \varphi[s]$, indicating that the assignment s satisfies formula φ in structure $\mathfrak{A}$, is defined recursively by:

1. $\mathfrak{A} \models Rt_1 \dots t_n[s] \iff (\bar{s}(t_1), \dots, \bar{s}(t_n)) \in R^{\mathfrak{A}}$.
2. $\mathfrak{A} \models = t_1 t_2[s] \iff \bar{s}(t_1) = \bar{s}(t_2)$.
3. $\mathfrak{A} \models \neg \psi[s] \iff \mathfrak{A} \not\models \psi[s]$.
4. $\mathfrak{A} \models \wedge \psi \xi[s] \iff \mathfrak{A} \models \psi[s] \wedge \mathfrak{A} \models \xi[s]$.
5. $\mathfrak{A} \models \vee \psi \xi[s] \iff \mathfrak{A} \models \psi[s] \vee \mathfrak{A} \models \xi[s]$.
6. $\mathfrak{A} \models \rightarrow \psi \xi[s] \iff \mathfrak{A} \not\models \psi[s] \vee \mathfrak{A} \models \xi[s]$.
7. $\mathfrak{A} \models \leftrightarrow \psi \xi[s] \iff (\mathfrak{A} \models \psi[s] \iff \mathfrak{A} \models \xi[s])$.
8. $\mathfrak{A} \models \forall x \psi[s] \iff \forall t \in |\mathfrak{A}| \mathfrak{A} \models \psi[s_{x \mapsto t}]$.
9. $\mathfrak{A} \models \exists x \psi[s] \iff \exists t \in |\mathfrak{A}| \mathfrak{A} \models \psi[s_{x \mapsto t}]$.

Example 5.3.6 (Evaluating Satisfaction in a Model). To make this abstract definition of satisfaction concrete, let us evaluate the sentence $\varphi = \exists x(x < y)$ in the structure of natural numbers $\mathfrak{A} = (\mathbb{N}, <^{\mathfrak{A}})$, where $<^{\mathfrak{A}}$ is the standard strictly-less-than relation.

Let us choose an assignment s where $s(y) := 2$. We evaluate $\mathfrak{A} \models \exists x(x < y)[s]$:

1. By the rule for existential quantifiers, this holds if and only if there exists some $t \in \mathbb{N}$ such that $\mathfrak{A} \models x < y[s_{x \mapsto t}]$.
2. By the relation satisfaction rule, this is equivalent to $(\bar{s}_{x \mapsto t}(x), \bar{s}_{x \mapsto t}(y)) \in <^{\mathfrak{A}}$.
3. Evaluating the term valuations under the modified assignment:

$$\bar{s}_{x \mapsto t}(x) = t \quad \text{and} \quad \bar{s}_{x \mapsto t}(y) = s(y) = 2$$

4. Thus, we need to find some $t \in \mathbb{N}$ such that $t < 2$. Choosing $t = 1$ (since $1 \in \mathbb{N}$ and $1 < 2$), the condition is satisfied.

Therefore, $\mathfrak{A} \models \exists x(x < y)[s]$ is true.

Example 5.3.7. We can write a structure as a tuple containing the universe and the interpreted symbols. For instance, the structure over the integers with addition and ordering is written $(\mathbb{Z}, +, \leq)$.

Theorem 5.3.8. Let φ be a formula. If two assignments s and r agree on all free variables of φ , then $\mathfrak{A} \models \varphi[s] \iff \mathfrak{A} \models \varphi[r]$.

Exercise 5.5. Prove Theorem 5.3.8 by structural induction.

Definition 5.3.9. A formula φ is a sentence if $\mathbf{FV}(\varphi) = \emptyset$.

Remark 5.3.10. Theorem 5.3.8 guarantees that for sentences, the choice of assignment has no effect on satisfaction.

Definition 5.3.11. Motivated by Theorem 5.3.8, we define $\mathfrak{A} \models \varphi$ to mean that $\mathfrak{A} \models \varphi[s]$ for every assignment s .

Definition 5.3.12. A formula φ is

- *satisfiable*, if $\mathfrak{A} \models \varphi[s]$ for some structure $\mathfrak{A}$ and assignment s .
- *unsatisfiable* (or a *contradiction*), if it is not satisfiable
- *valid* (or a *tautology*), written $\models \varphi$, if $\mathfrak{A} \models \varphi$ for every structure $\mathfrak{A}$
- *invalid*, written $\not\models \varphi$, if $\mathfrak{A} \not\models \varphi$ for some structure $\mathfrak{A}$

Definition 5.3.13. A structure $\mathfrak{A}$ is a model of a set of formulae Γ , written $\mathfrak{A} \models \Gamma$, if $\mathfrak{A} \models \varphi$ for every $\varphi \in \Gamma$.

Definition 5.3.14. We say Σ entails φ , written $\Sigma \models \varphi$, if for every structure $\mathfrak{A}$ and assignment s , if $\mathfrak{A} \models \psi[s]$ for all $\psi \in \Sigma$, then $\mathfrak{A} \models \varphi[s]$.

Definition 5.3.15. A set of formulae Σ is

- *satisfiable* if $\mathfrak{A} \models \Sigma$ for some structure $\mathfrak{A}$
- *unsatisfiable* if it is not satisfiable

Example 5.3.16. $(\mathbb{N}, +) \models \exists x \forall y x + y = y$.

Proof. Choose any assignment s . We show $(\mathbb{N}, +) \models \exists x \forall y x + y = y[s]$. This holds if there exists some $t \in \mathbb{N}$ such that $(\mathbb{N}, +) \models \forall y x + y = y[s_{x \mapsto t}]$. Choosing $t = 0$, we must show that for any $u \in \mathbb{N}$, we have $\bar{s}_{x \mapsto 0, y \mapsto u}(x) +^{\mathfrak{A}} \bar{s}_{x \mapsto 0, y \mapsto u}(y) = \bar{s}_{x \mapsto 0, y \mapsto u}(y)$, which reduces to $0 + u = u$. This is true. **Q.E.D.**

Exercise 5.6. Prove that $\forall x P(x) \models \exists x P(x)$. *Hint:* Review the definition of a structure carefully.

Exercise 5.7. Prove that $\exists xP(x) \models \neg\forall x\neg P(x)$ and $\neg\forall x\neg P(x) \models \exists xP(x)$.

5.4 Natural Deduction for First-Order Logic

To extend propositional Natural Deduction to First-Order Logic, we introduce rules for quantifiers and equality.

Symbol	Introduction Rules	Elimination Rules
$\forall$	$\frac{\phi(a)}{\forall x\phi(x)} \forall I^*$	$\frac{\forall x\phi(x)}{\phi(t)} \forall E$
$\exists$	$\frac{\phi(t)}{\exists x\phi(x)} \exists I$	$\frac{\begin{array}{c} \exists x\phi(x) \\ \vdots \\ \chi \end{array}}{\chi} \exists E_n^{**}$
$=$	$\frac{}{t = t} = I$	$\frac{t_1 = t_2 \quad \phi(t_1)}{\phi(t_2)} = E$

* Provided that a does not occur in any undischarged assumption upon which $\phi(a)$ depends.

** Provided that a does not occur in $\exists x\phi(x)$, in χ , or in any undischarged assumption other than $\phi(a)$.

Theorem 5.4.1 (First-Order Soundness). *Let $\Sigma \subseteq \mathbf{Frm}(\sigma)$ and $\varphi \in \mathbf{Frm}(\sigma)$. If $\Sigma \vdash \varphi$ in first-order Natural Deduction, then $\Sigma \models \varphi$.*

Exercise 5.8. Prove this theorem. *Hint:* Use induction on the height of proof trees.

Chapter 6

Metatheory of Classical Logic

6.1 Metatheory of Propositional Logic

We now examine the relationship between syntax and semantics, showing that our propositional proof systems are both sound and complete.

Theorem 6.1.1 (Tautology Soundness). *In the tautology system, $\Sigma \vdash \phi \implies \Sigma \models \phi$.*

Proof. Let $\varphi(n)$ be the statement: “The thesis holds for all proofs of length $k \leq n$ ”. We proceed by induction on n . The base case $\varphi(0)$ holds vacuously. Assume $\varphi(n)$. If the $(n+1)$ -th formula P_{n+1} is a tautology, then $\Sigma \models P_{n+1}$. If $P_{n+1} \in \Sigma$ is an assumption, then trivially $\Sigma \models P_{n+1}$. If P_{n+1} is obtained by Modus Ponens from P_i and $P_j = P_i \rightarrow P_{n+1}$, then by the inductive hypothesis $\Sigma \models P_i$ and $\Sigma \models P_i \rightarrow P_{n+1}$, which implies $\Sigma \models P_{n+1}$. **Q.E.D.**

We now prove the Compactness Theorem, which connects finite satisfiability with infinite satisfiability.

Definition 6.1.2. A set of formulae Σ is *finitely satisfiable* if every finite subset $\Sigma_0 \subseteq \Sigma$ is satisfiable.

Theorem 6.1.3 (Compactness). *A set of formulae Σ is satisfiable if and only if it is finitely satisfiable.*

Proof. ($\implies$) Trivial.

($\impliedby$) Let $\phi_0, \phi_1, \dots$ be an enumeration of all propositional formulae. Define:

$$\Delta_0 := \Sigma, \quad \Delta_{n+1} := \begin{cases} \Delta_n \cup \{\phi_n\} & \text{if finitely satisfiable,} \\ \Delta_n \cup \{\neg\phi_n\} & \text{otherwise.} \end{cases}$$

Let $\Delta := \bigcup_{n \in \mathbb{N}} \Delta_n$. Then Δ is finitely satisfiable. For any variable p , we define $\mathcal{I}(p) = \text{True} \iff p \in \Delta$. Using structural induction, one can show that the extended interpretation $\hat{\mathcal{I}}$ satisfies Δ , and thus satisfies $\Sigma \subseteq \Delta$. **Q.E.D.**

Exercise 6.1. ($\star$) Prove that if Δ_n is finitely satisfiable, then so is Δ_{n+1} in the Compactness proof.

Exercise 6.2. Prove that Δ is finitely satisfiable.

Exercise 6.3. $(\star)$ Prove that $\hat{\mathcal{L}}$ satisfies Δ by structural induction on formulae.

Exercise 6.4. Prove that the Compactness Theorem is equivalent to: “ $\Sigma \models \phi$ if and only if there exists a finite subset $\Sigma_0 \subseteq \Sigma$ such that $\Sigma_0 \models \phi$ ”.

Theorem 6.1.4 (Tautology Completeness). *In the tautology system, $\Sigma \models \phi \implies \Sigma \vdash \phi$.*

Proof. By Compactness, $\Sigma \models \phi$ implies there is a finite subset $\Sigma_0 = \{\psi_0, \dots, \psi_n\} \subseteq \Sigma$ such that $\Sigma_0 \models \phi$. By the propositional deduction theorem, this is equivalent to $\vdash \psi_0 \rightarrow \psi_1 \rightarrow \dots \rightarrow \phi$. Since this is a semantic tautology, it is an axiom of the tautology system. Modus Ponens then yields a proof of ϕ from Σ_0 . **Q.E.D.**

Exercise 6.5. Prove that $\Gamma \vdash \phi \iff \Gamma \cup \{\neg\phi\} \vdash \perp$ in the tautology system.

6.2 Completeness of First-Order Logic

We now prove Gödel’s Completeness Theorem for first-order logic, showing that syntactic consistency implies semantic satisfiability.

Theorem 6.2.1. *Let Γ be consistent over $\mathcal{L}$. If $c_0, c_1, \dots \notin \mathcal{L}$, then Γ is consistent over $\mathcal{L} \cup \{c_0, c_1, \dots\}$.*

Proof. Suppose otherwise. Then there is a derivation of $\perp$. Since formal proofs are finite, this derivation uses only a finite subset of the new constants. Replacing each constant with a variable not occurring in the proof yields a valid proof of $\perp$ using only formulae from $\mathcal{L}$, contradicting the consistency of Γ . **Q.E.D.**

Theorem 6.2.2 (Henkin Constants). *Any consistent set Γ can be extended to a consistent set $\Gamma \cup \{\neg\forall x\phi_n(x) \rightarrow \neg\phi_n(c_{k_n}) \mid n \in \mathbb{N}\}$, where $\phi_0, \dots$ is an enumeration of all FOL formulae and c_{k_n} is a sequence of constants where c_{k_n} does not occur in any ϕ_m for $m < n$.*

Proof. Let $\Gamma'_0 := \Gamma$, and $\Gamma'_{n+1} := \Gamma'_n \cup \{\neg\forall x\phi_n(x) \rightarrow \neg\phi_n(c_{k_n})\}$. Let $\varphi(n)$ be the statement: “ Γ'_n is consistent”. Assume $\neg\varphi(n+1)$. Then $\Gamma'_n \cup \{\neg\forall x\phi_n(x) \rightarrow \neg\phi_n(c_{k_n})\} \vdash \perp$, which implies $\Gamma'_n \vdash \neg\forall x\phi_n(x) \wedge \phi_n(c_{k_n})$. This yields $\Gamma'_n \vdash \neg\forall x\phi_n(x)$ and $\Gamma'_n \vdash \forall x\phi_n(x)$ (by generalization), which means $\neg\varphi(n)$. By contraposition, $\varphi(n) \implies \varphi(n+1)$. Let $\Gamma' := \bigcup_{n \in \mathbb{N}} \Gamma'_n$. Since any formal proof of $\perp$ from Γ' would be finite and thus contained in some Γ'_n , consistency of all Γ'_n guarantees that Γ' is consistent. **Q.E.D.**

Definition 6.2.3. A set of formulae T has the *Henkin property* (or the *witness property*) if whenever $T \vdash \exists x \varphi(x)$, then $T \vdash \varphi(c)$ for some constant symbol c .

Remark 6.2.4. The set of formulae yielded by Theorem 6.2.2 has the Henkin property.

Definition 6.2.5. A set of formulae Γ is maximally consistent if it is consistent and for every formula ϕ , either $\phi \in \Gamma$ or $\neg\phi \in \Gamma$.

Theorem 6.2.6 (Lindenbaum's Lemma). *A consistent set Γ can be extended to a maximally consistent set Δ .*

Proof. Let $\phi_0, \phi_1, \dots$ be an enumeration of all FOL formulae. Define:

$$\Delta_0 := \Gamma, \quad \Delta_{n+1} := \begin{cases} \Delta_n \cup \{\phi_n\} & \text{if consistent,} \\ \Delta_n \cup \{\neg\phi_n\} & \text{otherwise.} \end{cases}$$

As in the proof of Compactness, each Δ_n is consistent, and thus their union $\Delta := \bigcup_{n \in \mathbb{N}} \Delta_n$ is consistent and complete. **Q.E.D.**

Exercise 6.6. Verify that if $\Delta_n \cup \{\phi_n\} \vdash \perp$ and $\Delta_n \cup \{\neg\phi_n\} \vdash \perp$, then $\Delta_n \vdash \perp$.

Exercise 6.7. Verify that $\Delta \vdash \perp \implies \Delta_n \vdash \perp$ for some n .

Definition 6.2.7. Let Δ be a complete consistent set containing Henkin constants. A Γ -verifier is a structure $\mathfrak{A}$ where $|\mathfrak{A}| := \mathbf{Trm}(\mathcal{L} \cup \{c_0, \dots\})$ defined by:

1. $R^{\mathfrak{A}}(t_1, \dots, t_n) \iff R(t_1, \dots, t_n) \in \Delta$.
2. $f^{\mathfrak{A}}(t_1, \dots, t_n) := f(t_1, \dots, t_n)$.
3. $E^{\mathfrak{A}}(t_1, t_2) \iff t_1 = t_2 \in \Delta$.

The valuation s is defined by $s(x) := x$ for every variable x .

Exercise 6.8. Prove that $\bar{s}(t) = t$ for every term t .

Theorem 6.2.8. $\mathfrak{A} \models \phi^* \iff \phi \in \Delta$, where ϕ^* is the formula obtained by replacing every instance of $t_1 = t_2$ with $E(t_1, t_2)$.

Proof. We proceed by induction on formulae. We illustrate the case for quantifiers: $\mathfrak{A} \models (\forall x \phi(x))^* \iff \forall t \in |\mathfrak{A}| \mathfrak{A} \models \phi^*[s_{x \mapsto t}] \iff \forall t \in |\mathfrak{A}| \phi(t) \in \Delta$ (by inductive hypothesis). If $\neg \forall x \phi(x) \in \Delta$, then by Henkin properties, $\neg \phi(c_{k_n}) \in \Delta$ for some constant c_{k_n} , which contradicts consistency. **Q.E.D.**

Exercise 6.9. Complete the remaining inductive cases of the proof.

Definition 6.2.9. A Γ -model is the quotient structure $\mathfrak{A}' := \mathfrak{A}/E^{\mathfrak{A}}$ defined by:

1. $R^{\mathfrak{A}'}([t_1], \dots, [t_n]) \iff R^{\mathfrak{A}}(t_1, \dots, t_n)$.
2. $f^{\mathfrak{A}'}([t_1], \dots, [t_n]) := [f^{\mathfrak{A}}(t_1, \dots, t_n)]$.

Exercise 6.10. Prove that $\mathfrak{A}' \models \phi \iff \phi \in \Delta$. Conclude that $\mathfrak{A}' \models \Delta$.

Theorem 6.2.10 (Completeness Theorem). *If Γ is consistent, then Γ has a model. Consequently, $\Gamma \models \phi \implies \Gamma \vdash \phi$.*

Proof. Let Γ be consistent. By the Henkin construction and Lindenbaum's lemma, we extend Γ to a complete consistent set Δ containing Henkin constants. The Γ -model $\mathfrak{A}'$

satisfies Δ , and since $\Gamma \subseteq \Delta$, we have $\mathfrak{A}' \models \Gamma$. Now, if $\Gamma \models \phi$, then $\Gamma \cup \{\neg\phi\}$ has no model and is thus inconsistent. Therefore, $\Gamma \cup \{\neg\phi\} \vdash \perp$, which implies $\Gamma \vdash \phi$. **Q.E.D.**

Remark 6.2.11. The proofs above involve are quite involved. The big picture is this:

1. Take a consistent set Γ .
2. Extend Γ by Henkin constants, yielding Γ' .
3. Extend Γ' to a maximal consistent set Δ .
4. Find a model $\mathfrak{A}$ of Δ by using the terms in the language as the universe.
5. From $\mathfrak{A}$ make $\mathfrak{A}'$, a structure that associates two terms t_1, t_2 iff $t_1 = t_2 \in \Delta$.
6. Because Γ was arbitrary, any consistent set is satisfiable, which makes the proof system complete.

Commentary. The construction of the model might seem unmotivated, but it mirrors how we conceptualise unknown entities through their descriptions alone: we might read of Achilles and Hector, and where the Iliad turns supernatural or strains against physical possibility, we find no trouble reasoning about the narrative, because it is internally consistent. Some of what we picture may be impossible to visualise properly, yet we accept it anyway, and it lives in our minds simply as words about a world. This is exactly how we build a model here: its objects terms and statements of our language, held together just by their own consistency.

6.3 Completeness Checklist

The entire metatheoretic proof depended only on a few syntactic properties of the consequence relation ($\vdash$), not the proof system itself.

Let $\Gamma, \Delta \subseteq \mathbf{Frm}(\sigma)$ be sets of formulae, let φ, ψ be formulae. Let $\vdash$ satisfy:

- I. Structural Properties (Tarskian Consequence)**
 1. **Reflexivity:** If $\varphi \in \Gamma$, then $\Gamma \vdash \varphi$.
 2. **Monotonicity:** If $\Gamma \vdash \varphi$ and $\Gamma \subseteq \Delta$, then $\Delta \vdash \varphi$.
 3. **Transitivity (Cut):** If $\Gamma \vdash \psi$ for all $\psi \in \Delta$, and $\Delta \vdash \varphi$, then $\Gamma \vdash \varphi$.
 4. **Finitariness (Compactness):** If $\Gamma \vdash \varphi$, there exists a finite subset $\Gamma_0 \subseteq \Gamma$ such that $\Gamma_0 \vdash \varphi$. (This is necessary to show that the union of a chain of consistent sets remains consistent).
- II. Propositional Connective Properties**
 5. **Deduction Theorem:** $\Gamma \cup \{\psi\} \vdash \varphi \iff \Gamma \vdash \psi \rightarrow \varphi$.
 6. **Double Negation Elimination:** If $\Gamma \cup \{\neg\varphi\} \vdash \perp$, then $\Gamma \vdash \varphi$. (This classical property forces either φ or $\neg\varphi$ into our complete consistent set).
 7. **Ex Falso Quodlibet:** $\{\perp\} \vdash \varphi$ for any arbitrary formula φ .
- III. Quantifier Properties**
 8. **Universal Instantiation:** $\{\forall x\varphi(x)\} \vdash \varphi(t)$ for any term t free for x in $\varphi(x)$.

9. **Universal Generalization:** If $\Gamma \vdash \varphi(c)$ and the constant symbol c does not occur in Γ or $\varphi(x)$, then $\Gamma \vdash \forall x\varphi(x)$. (This is the engine that drives the creation of Henkin constants).

10. **Existential Duality:** $\exists x\varphi(x) \vdash \neg\forall x\neg\varphi(x)$ and $\forall x\varphi(x) \vdash \neg\exists x\neg\varphi(x)$.

IV. Equality Properties (if $=$ is in the signature) 11. **Reflexivity of Equality:** $\vdash t = t$ for any term t .

12. **Substitution (Leibniz's Law):** $\{t_1 = t_2, \varphi(t_1)\} \vdash \varphi(t_2)$ for any atomic formula φ . (This ensures that equality is a congruence relation, allowing us to build the well-defined quotient structure $\mathfrak{A}'$).

Then the proof system that $\vdash$ denotes is complete.

Chapter 7

Axiomatic Set Theory

7.1 Zermelo-Fraenkel Axioms

As shown by Russell's paradox, naive set theory is inconsistent. We must therefore build set theory axiomatically using First-Order Logic.

Definition 7.1.1. The language of Set Theory contains a single binary relation symbol $\in$.

Theorem 7.1.2. *Naive Set Theory (NST) is inconsistent and thus has no model.*

Proof. Assume a model exists. By unrestricted comprehension, we can define the set $A := \{x \mid x \notin x\}$. Then $A \in A \iff A \notin A$, which is a logical contradiction. Thus, $\text{NST} \models \perp$. **Q.E.D.**

To avoid this inconsistency, we restrict how sets are constructed.

Axiom 1 (Extensionality). $\mathbf{Ex} := \forall x \forall y (\forall z (z \in x \iff z \in y) \implies x = y)$.

Intuition. Sets are uniquely determined by their elements.

Commentary. This axiom formalizes Leibniz's law (the Identity of Indiscernibles): if two objects cannot be distinguished by any property, they must be identical. Since $\in$ is our only primitive relation, sets with the same members are completely indistinguishable.

Exercise 7.1. Prove that $\mathbf{Ex} \vdash \forall x \forall y (\forall z (z \in x \iff z \in y) \iff x = y)$.

Axiom 2 (Empty Set). $\mathbf{ES} := \exists x \forall y \neg(y \in x)$.

Exercise 7.2. Prove that $\{\mathbf{Ex}, \mathbf{ES}\} \vdash \exists! x \forall y \neg(y \in x)$. We denote this unique set by $\emptyset$.

Axiom 3 (Pair). $\mathbf{Pa} := \forall x \forall y \exists z \forall w (w \in z \iff (w = x \vee w = y))$.

Axiom 4 (Union). $\mathbf{U} := \forall x \exists y \forall z (z \in y \iff \exists w (z \in w \wedge w \in x))$.

Exercise 7.3. Prove that triples can be defined in $\{\mathbf{Ex}, \mathbf{ES}, \mathbf{Pa}, \mathbf{U}\}$.

Axiom 5 (Specification / Restricted Comprehension). *For any formula $\varphi(z)$, the following is an axiom: $\forall x \exists y \forall z (z \in y \iff (z \in x \wedge \varphi(z)))$. We call the collection of such axioms $\mathbf{S}$.*

Axiom 6 (Replacement). *For any functional in z formula $\varphi(w, z)$, the following is an axiom: $\forall x \exists y \forall z (z \in y \iff \exists w (w \in x \wedge \varphi(w, z)))$. We call the collection of such axioms $\mathbf{R}$.*

Remark 7.1.3. Specification and Replacement are axiom schemas, meaning they represent infinitely many individual axioms (one for each formula φ).

Exercise 7.4. Prove that $\{\mathbf{Ex}, \mathbf{ES}, \mathbf{R}\} \vdash \mathbf{S}$.

Axiom 7 (Power Set). $\mathbf{PS} := \forall x \exists y \forall z (z \in y \iff \forall w (w \in z \rightarrow w \in x))$.

Definition 7.1.4 (Kuratowski's definition of an ordered pair). The ordered pair (a, b) is formally defined set-theoretically as: $(a, b) := \{\{a\}, \{a, b\}\}$.

Exercise 7.5. Verify that Kuratowski's definition of an ordered pair satisfies the characteristic property: $(a, b) = (c, d) \iff a = c \wedge b = d$.

Exercise 7.6. Write down a formal formula ψ stating that for any a, b , the ordered pair (a, b) exists, and prove $\{\mathbf{Ex}, \mathbf{Pa}\} \vdash \psi$.

Commentary. Now is the time to think about how to define the natural numbers. Similarly to how a metre is defined by declaring some arbitrary object to be a metre long and then comparing everything else to it, we could find an example set with n elements and declare that a set has n elements if there is a bijection between it and our example set. Thus, to define n , we just need to decide on which n elements it should contain. Without much thought, we can already define $0 := \emptyset$. It is clean and elegant to define as n the set of natural numbers smaller than it: $1 = \{0\}, 2 = \{0, 1\}, \dots$

Definition 7.1.5. We define the successor of a set n as $n + 1 := n \cup \{n\}$.

Axiom 8 (Infinity). $\mathbf{I} := \exists x (\emptyset \in x \wedge \forall y (y \in x \rightarrow y + 1 \in x))$.

Exercise 7.7. Prove that $\{\mathbf{Ex}, \mathbf{I}\} \cup \mathbf{R} \vdash \mathbf{ES}$. This shows the empty set axiom is redundant.

Definition 7.1.6. We define $Z^- := \{\mathbf{Ex}, \mathbf{ES}, \mathbf{Pa}, \mathbf{U}, \mathbf{PS}, \mathbf{I}\}$, and $ZF^- := Z^- \cup \mathbf{R}$.

Definition 7.1.7. Let $\mathcal{I}$ be an inductive set guaranteed by $\mathbf{I}$. We define: $\omega := \bigcap \{x \in \mathcal{P}(\mathcal{I}) \mid x \text{ is inductive}\}$.

Exercise 7.8. Let S be an inductive set. Prove that $\omega \subseteq S$.

Definition 7.1.8. We define the set of natural numbers as $\mathbb{N} := \omega$.

Theorem 7.1.9 (Induction on ω). *Let $S \subseteq \omega$. If $0 \in S$ and $\forall n (n \in S \rightarrow n + 1 \in S)$, then $S = \omega$.*

Proof. By assumption, S is inductive. Thus, $\omega \subseteq S$. Since $S \subseteq \omega$, we have $S = \omega$. **Q.E.D.**

Exercise 7.9. Define addition recursively on ω and prove that addition is associative.

Remark 7.1.10. All of arithmetic behaves as it should in ZF^- . The reader, who is less swayed by life's distractions than the author, is welcome to state and prove some of the theorems himself.

Definition 7.1.11. A set X is *transitive* if $\forall x \in X \ x \subseteq X$.

Exercise 7.10. $(\star)$ Prove that X is transitive if and only if $X \subseteq \mathcal{P}(X)$. Prove that there exists no set X such that $\mathcal{P}(X) \subseteq X$, and conclude that X is transitive if and only if $X \subset \mathcal{P}(X)$.

7.2 Classes

The axioms of ZF^- place restrictions on what is and is not a set. In order not to let ourselves become too acquainted with the theory (for doing so has its downsides), we decide that we still wish to talk and think about "collections" that might not be sets.

Definition 7.2.1. For any formula $\varphi(x, \bar{t})$, we define a class $A = \{x \mid \varphi(x, \bar{t})\}$ of elements x such that $\varphi(x, \bar{t})$. For any such x , we write $x \in A$.

Definition 7.2.2. A class that is not a set is called a proper class.

Definition 7.2.3. We define the class of all sets $\mathbf{V} := \{x \mid x = x\}$.

Definition 7.2.4. We generalise the notations for sets to classes in a straightforward manner:

- $A \subseteq B \iff \forall x(x \in A \rightarrow x \in B)$
- $\{x \mid \varphi(x, \bar{t})\} \cap \{x \mid \psi(\bar{x}, \bar{s})\} := \{x \mid \varphi(x, \bar{t}) \wedge \psi(x, \bar{s})\}$
- $\{x \mid \varphi(x, \bar{t})\} \cup \{x \mid \psi(\bar{x}, \bar{s})\} := \{x \mid \varphi(x, \bar{t}) \vee \psi(x, \bar{s})\}$
- $\bigcap \{x \mid \varphi(x, \bar{t})\} := \{z \mid \forall x(\varphi(x, \bar{t}) \wedge z \in x)\}$
- $\bigcup \{x \mid \varphi(x, \bar{t})\} := \{z \mid \exists x(\varphi(x, \bar{t}) \wedge z \in x)\}$
- $\{x \mid \varphi(x, \bar{t})\}^c := \{x \mid \neg \varphi(x, \bar{t})\}$

Remark 7.2.5. Note that we don't have to generalise the notion of relations or functions, because we've only defined set representations of functions and relations. As such, relations and functions were never constrained by having to be sets.

Commentary. Notice that in the set theories we are considering, the variables of our language range exclusively over sets; no other objects reside in our universe. Classes, then, do not exist. That is not alarming — for the mind is itself thin and wonderfully easy to move, as Lucretius said. It is also important to recognise that it is grammatically nonsensical to write, for a proper class A , $A \in B$ or $\{A\}$. Proper classes are sometimes said to be “too large” to fit in sets. This, as we shall see, is not just a saying, but a precise diagnosis.

7.3 Ordinal Numbers and Order Isomorphisms

We now apply our axiomatic set theory to order theory, culminating in the study of ordinals.

Theorem 7.3.1 (Well-Ordered Induction). *Let $(X, \leq)$ be a well-ordered set and let Π be a property. If $\Pi(a)$ holds whenever $\Pi(y)$ holds for all $y < a$, then $\Pi(x)$ holds for all $x \in X$.*

Proof. Let $S := \{x \in X \mid \neg \Pi(x)\}$. If $S \neq \emptyset$, let $a := \min_{\leq} S$. Then $\Pi(y)$ holds for all $y < a$, which implies $\Pi(a)$ must hold, a contradiction. Thus, $S = \emptyset$. **Q.E.D.**

Definition 7.3.2. Let $(X, \leq)$ and $(Y, \hat{\leq})$ be partially ordered sets. A function $f: X \rightarrow Y$ is an *order isomorphism* if f is bijective and $\forall x, y \in X$ ($x \leq y \iff f(x) \hat{\leq} f(y)$). We write $(X, \leq) \cong_{\text{Ord}} (Y, \hat{\leq})$.

Definition 7.3.3. Let $(X, \leq)$ and $(Y, \hat{\leq})$ be partially ordered sets and $f: X \rightarrow Y$. We call f *order-preserving* if $\forall x, x' \in X$ ($x < x' \rightarrow f(x) < f(x')$).

Exercise 7.11. Let $(X, \leq)$ be a totally ordered set, $(Y, \hat{\leq})$ be a partially ordered set and $f: X \rightarrow Y$ be a bijection. Prove that f is an order isomorphism if and only if f is order-preserving. Show that the assumption that $(X, \leq)$ be a totally ordered set is necessary.

Exercise 7.12. Prove that $\cong_{\text{Ord}}$ is an equivalence relation on partially ordered sets.

Definition 7.3.4. An order isomorphism from a partially ordered set to itself is called an *order automorphism*.

Theorem 7.3.5. *Let $(X, \leq)$ be a well-ordered set and $f: X \rightarrow X$ an order automorphism. Then $\forall x \in X$ $x \leq f(x)$.*

Proof. Let $S := \{x \in X \mid f(x) < x\}$. If $S \neq \emptyset$, let $s := \min_{\leq} S$. Since $f(s) < s$, applying f yields $f(f(s)) < f(s)$, which contradicts the minimality of s . Thus, $S = \emptyset$. **Q.E.D.**

Theorem 7.3.6. *The only order automorphism of a well-ordered set $(X, \leq)$ is the identity map id_X .*

Proof. If f is an automorphism, then f^{-1} is also an automorphism. Thus, $\forall x \in X$, we have $x \leq f(x)$ and $x \leq f^{-1}(x)$, which implies $f(x) \leq x$. Hence, $f(x) = x$. **Q.E.D.**

Exercise 7.13. Verify that id_X is an automorphism, and if f is an automorphism, then f^{-1} is also an automorphism.

Exercise 7.14. Prove that if two well-ordered sets are isomorphic, their isomorphism is unique.

Theorem 7.3.7. *A well-ordered set $(X, \leq)$ is never isomorphic to any of its proper initial segments X_a .*

Proof. Suppose $f: X \rightarrow X_a$ is an isomorphism. Then $f(a) \in X_a$, which means $f(a) < a$. This contradicts the fact that $x \leq f(x)$ for all $x \in X$. **Q.E.D.**

Definition 7.3.8. Let $(X, \leq)$ and $(Y, \hat{\leq})$ be partially ordered sets and $(X, \leq)$ be isomorphic to an initial segment of $(Y, \hat{\leq})$. We then write $(X, \leq) \sqsubseteq (Y, \hat{\leq})$.

Theorem 7.3.9. *Let $X := (X, \leq)$ and $Y := (Y, \hat{\leq})$ be well-ordered sets. Then exactly one of the following holds:*

1. $X \sqsubset Y$, or
2. $X \cong_{\text{Ord}} Y$, or
3. $Y \sqsubset X$.

Proof. Define $f := \{(x, y) \in X \times Y \mid X_x \cong_{\text{Ord}} Y_y\}$. By Theorem 7.3.7, f is a functional, injective, order-preserving relation.

Notice that $\text{dom } f$ and $\text{ran } f$ are initial segments of X and Y . If $\text{dom } f \neq X$ and $\text{ran } f \neq Y$, let $x_0 := \min(X \setminus \text{dom } f)$ and $y_0 := \min(Y \setminus \text{ran } f)$. Then:

$$\text{dom } f = X_{x_0} \quad \text{and} \quad \text{ran } f = Y_{y_0} \implies X_{x_0} \cong_{\text{Ord}} Y_{y_0} \implies (x_0, y_0) \in f,$$

contradicting $x_0 \notin \text{dom } f$. Thus, $\text{dom } f = X$ or $\text{ran } f = Y$:

$$\begin{aligned} \text{dom } f = X \wedge \text{ran } f = Y &\implies X \cong_{\text{Ord}} Y, \\ \text{dom } f = X \wedge \text{ran } f = Y_{y_0} &\implies X \cong_{\text{Ord}} Y_{y_0} \implies X \sqsubset Y, \\ \text{dom } f = X_{x_0} \wedge \text{ran } f = Y &\implies X_{x_0} \cong_{\text{Ord}} Y \implies Y \sqsubset X. \end{aligned} \quad \textbf{Q.E.D.}$$

Exercise 7.15. Prove that the three conditions in Theorem 7.3.9 are mutually exclusive.

Definition 7.3.10. Let $(X, \leq)$ be a well-ordered set. Then X is an ordinal if $\forall \alpha \in X \ X_\alpha = \alpha$.

Exercise 7.16. Prove that ordinals are transitive (as sets).

Theorem 7.3.11. *Let $(X, \leq)$ be a well-ordered set, X be an ordinal and $Y \in X$. Then $(Y, \leq|_{Y \times Y})$ is an ordinal.*

Proof. $(Y, \leq|_{Y \times Y})$ is a well-ordered set, since $Y = X_Y$. Pick any $\alpha \in Y$. Then $Y_\alpha = \{y \in Y \mid y < \alpha\} = \{y \in X_Y \mid y < \alpha\} = \{y \in X \mid y < \alpha\}$, since X is well-ordered. **Q.E.D.**

Theorem 7.3.12. *Let X be an ordinal and $\alpha, \beta \in X$. Then the following assertions are pairwise equivalent:*

1. $\alpha < \beta$,
2. $\alpha \in \beta$,
3. $\alpha \subset \beta$,
4. $\alpha \sqsubset \beta$.

Proof. We prove $(1) \iff (2) \implies (4) \implies (3) \implies (1)$:

(1) $\iff$ (2): $\alpha < \beta \iff \alpha \in X_\beta \iff \alpha \in \beta$ (since $X_\beta = \beta$).

(2) $\implies$ (4): $\alpha \in \beta \implies \alpha = X_\alpha = \beta_\alpha \implies \alpha \sqsubset \beta$.

(4) $\implies$ (3): $\alpha \sqsubset \beta \implies \exists \gamma \in \beta \ \alpha = \beta_\gamma \implies \alpha \subset \beta$.

(3) $\implies$ (1): $\alpha \subset \beta \implies \alpha \neq \beta \wedge \beta \not\subset \alpha \implies \neg(\alpha = \beta) \wedge \neg(\beta < \alpha) \implies \alpha < \beta$ (by totality of $<$).

Q.E.D.

Remark 7.3.13. It follows from Theorem 7.3.12 that if α is an ordinal, then $\in$ treated as a relation on $\alpha \times \alpha$ is well-founded.

Definition 7.3.14. Following Theorem 7.3.12, we define for any two ordinals α, β , $\alpha < \beta$ to mean $\alpha \in \beta$.

Theorem 7.3.15. *Let α, β be ordinals. Then $\alpha \subset \beta \implies \alpha \in \beta$.*

Proof. Let γ be the minimal element of $\beta \setminus \alpha$. Then $\gamma \subseteq \alpha$. Pick any $\delta \in \alpha$. Then $\gamma \leq \delta$ or $\delta < \gamma$. If $\gamma \leq \delta$, then $\gamma \in \delta$ or $\gamma = \delta$, both of which lead to $\gamma \in \alpha$, a contradiction. Therefore, we get that $\delta \in \gamma$ and so $\alpha \subseteq \gamma$, so $\alpha = \gamma$, meaning that $\alpha \in \beta$. **Q.E.D.**

Remark 7.3.16. It follows from Exercise 7.16, Theorem 7.3.15 and that $\alpha \in \beta \implies \alpha = \beta_\alpha$ that $\alpha < \beta$, $\alpha \in \beta$, $\alpha \subset \beta$, $\alpha \sqsubset \beta$ are pairwise equivalent.

Theorem 7.3.17. *Let α, β be ordinals, then $\alpha < \beta$ or $\alpha = \beta$ or $\beta < \alpha$.*

Proof. Suppose that $\beta \not\leq \alpha$. Since $\emptyset \neq \beta \setminus \alpha \subseteq \beta$, we may pick an $\in$ -minimal element γ in this set. We get that $\gamma \subseteq \alpha$, so $\gamma \in \alpha$ or $\gamma = \alpha$, but $\gamma \notin \alpha$, therefore $\alpha = \gamma \in \beta$. **Q.E.D.**

Exercise 7.17. Show that Theorem 7.3.17 follows from Theorem 7.3.9.

Definition 7.3.18. Let X be a set of ordinals. We define $\inf X := \bigcap X$ and $\sup X := \bigcup X$.

Definition 7.3.19. We define **Ord** := $\{\alpha \mid \alpha \text{ is an ordinal number}\}$.

Theorem 7.3.20. *Let α be an ordinal and X be a set of ordinals. Then:*

1. $\alpha = \{\beta \in \mathbf{Ord} \mid \beta < \alpha\}$
2. $\alpha + 1$ is an ordinal,
3. $\sup X$ is an ordinal that is the least upper bound of X in the ordinal numbers,
4. $\inf X$ is an ordinal that is the greatest lower bound of X in the ordinal numbers.
5. $\in$ treated as a relation on $\mathbf{Ord}$ is well-founded.

Exercise 7.18. Prove Theorem 7.3.20.

Remark 7.3.21. Theorem 7.3.20 shows that notation introduced in Definition 7.3.18 is in accordance with Definition 1.5.6.

Theorem 7.3.22 (Extended Burali-Forti). *Let A be a subclass of $\mathbf{Ord}$ that is not bounded above in $\mathbf{Ord}$. Then A is not a set.*

Proof. Suppose that A is a set. By theorem 7.3.20, $\sup A + 1 \in \mathbf{Ord}$, but there is no $a \in A$ that is greater than it, since $\sup A$ is an upper bound on A , a contradiction. **Q.E.D.**

Definition 7.3.23. Let $\alpha > 0$ be an ordinal. If there exists an ordinal β such that $\alpha = \beta + 1$, we call α a successor ordinal. Else, we call α a limit ordinal.

Theorem 7.3.24 (Transfinite Induction). *Let A be a class of ordinal numbers such that for any $\alpha \in A$ and limit ordinal λ :*

- **Base case:** $0 \in A$,
- **Successor case:** $\alpha + 1 \in A$
- **Limit case:** If $\forall \alpha < \lambda \alpha \in A$, then $\lambda \in A$,

then $A = \mathbf{Ord}$.

Proof. Any smallest counterexample, if it exists, must be of one of the three forms above. **Q.E.D.**

Theorem 7.3.25. *Let $(X, \hat{\leq})$ be a well ordered set. Then there exists a unique ordinal number α such that $(X, \hat{\leq}) \cong_{\mathbf{Ord}} (\alpha, \leq)$.*

Proof. Uniqueness follows from Theorem 7.3.6. Given a well-ordered set X , define $F(x) = \alpha$ if $(X, \leq)_x \cong_{\mathbf{Ord}} (\alpha, \in)$. If such α exists, it is unique. By **R**, $F(X)$ is a set. Such an α exists for every $x \in X$, otherwise consider the least x for which it does not. If $\gamma = \inf(\mathbf{Ord} \setminus F(X))$, then $F(X) = \gamma$ and $X \cong_{\mathbf{Ord}} \gamma$. **Q.E.D.**

Theorem 7.3.26 (Transfinite Recursion). *Let $G: \mathbf{V} \rightarrow \mathbf{V}$ be a function. There exists a unique class function $F: \mathbf{Ord} \rightarrow \mathbf{V}$ such that for every ordinal α :*

$$F(\alpha) = G(F|_{\alpha})$$

Lemma 7.3.27. *For any ordinal α , there is at most one function $f^\alpha: \alpha \rightarrow \mathbf{V}$ such that $\forall \beta < \alpha, f^\alpha(\beta) = G(f^\alpha|_{\beta})$. Furthermore, if $\beta < \alpha$, then $f^\alpha|_{\beta} = f^\beta$.*

Proof. Transfinite induction for $\varphi(\alpha) := \exists f^\alpha : \alpha \longrightarrow \mathbf{V} \forall \beta < \alpha f^\alpha(\beta) = G(f^\alpha|_\beta)$. We consider the limit case: let λ be a limit ordinal and $\varphi(\alpha)$ for all $\alpha < \lambda$. By the Axiom of Replacement, the collection $A := \{f^\alpha \mid \alpha < \lambda\}$ is a set. By Lemma 7.3.27, $(A, \subseteq)$ is a chain. Thus, $f^\lambda := \bigcup_{\alpha < \lambda} f^\alpha$ is a function with domain λ satisfying the condition.

We define the class function F as the union of all such partial functions:

$$F := \bigcup_{\alpha \in \mathbf{Ord}} f^\alpha$$

By Lemma 7.3.27, F is a well-defined class function. Uniqueness of F follows from the uniqueness of its constituent partial functions. **Q.E.D.**

Exercise 7.19. Prove Lemma 7.3.27.

Exercise 7.20. Prove that $\varphi(0)$ and for any α , $\varphi(\alpha) \rightarrow \varphi(\alpha+1)$ in the proof of Theorem 7.3.26, and justify that F , as defined, is given by the Axiom of Replacement.

7.4 Sequences of Ordinal Numbers

We now study ordinal number sequences much like real analysis studies real number sequences.

Definition 7.4.1. An α -sequence of elements of X is defined to be a function $\xi : \alpha \longrightarrow X$. Often $\xi(\beta)$ is denoted as ξ_β .

Definition 7.4.2. For a sequence $\xi : \alpha \longrightarrow X$ and a relation R , we define $\sup_{R(\beta)} \xi_\beta := \sup \xi(\{\beta < \alpha \mid R(\beta)\})$ and similarly $\inf_{R(\beta)} \xi_\beta$.

Definition 7.4.3. Let ξ be an α -sequence of ordinals. We say that ξ converges to η and write $\xi \rightarrow \eta$ or $\lim_{\beta < \alpha} \xi_\beta = \eta$, if $\forall \lambda < \eta \exists \beta \forall \gamma (\beta \leq \gamma < \alpha \rightarrow \lambda < \xi_\gamma \leq \eta)$.

Commentary. Definition 7.4.3 is essentially a rewriting of the definition of a limit of real numbers. Recall that a real number sequence a converges to L if and only if $\forall \varepsilon > 0 \exists N \forall n \geq N |a_n - L| < \varepsilon$. Both definitions make formal the same natural condition: for any boundary around the limit, the sequence eventually enters it and remains within it. This, in general, is how a limit of a sequence is always defined.

Exercise 7.21. Let ξ be an α -sequence of ordinals. Prove that ξ converges if and only if $\lim_{\beta < \alpha} \inf_{\beta < \gamma} \xi_\beta = \lim_{\beta < \alpha} \sup_{\beta < \gamma} \xi_\beta$.

7.5 Axiom of Choice

Axiom 9 (Choice). The axiom of choice, written **AC**, states that for any set of nonempty sets $\mathcal{A}$ there exists a function $f : \mathcal{A} \longrightarrow \bigcup \mathcal{A}$ such that $\forall A \in \mathcal{A} f(A) \in A$.

Exercise 7.22. Write **AC** as a FOL formula over the language of set theory.

Definition 7.5.1. We define ZFC^- as $\text{ZF}^- \cup \{\mathbf{AC}\}$.

Exercise 7.23. Let $\mathcal{A}$ be a set of nonempty subsets of a fixed well-ordered set. Prove that $\text{ZF}^- \vdash \exists f : \mathcal{A} \longrightarrow \bigcup \mathcal{A} \forall A \in \mathcal{A} f(A) \in A$.

Exercise 7.24. Let $\mathcal{A}$ be a finite set of nonempty sets. Prove that $\text{ZF}^- \vdash \exists f : \mathcal{A} \longrightarrow \bigcup \mathcal{A} \forall A \in \mathcal{A} f(A) \in A$. (Hint: induction over the size of $\mathcal{A}$).

Exercise 7.25. Show that, assuming ZFC^- , if there exists a surjection $f : B \longrightarrow A$, then there exists an injection $g : A \longrightarrow B$.

Theorem 7.5.2 (Hausdorff). *Let $(X, \leq)$ be a partially ordered set. Then there exists a maximal chain in X (i.e. a chain $\mathcal{C} \subseteq X$ that is a maximal element of the partially ordered set $(\text{Chains}(X), \subseteq)$).*

The theorem is usually referred to as “Hausdorff Maximal Principle” and denoted by **HP**.

Proof. From the axiom of choice we get a function $g : \mathcal{P}(X) \setminus \{\emptyset\} \longrightarrow X$. For any $A \subseteq X$ we define $E_A := \{x \in X \setminus A \mid A \cup \{x\} \text{ is a chain}\}$. We also define a function $f : \text{Chains}(X) \longrightarrow \text{Chains}(X)$ by $f(A) := \begin{cases} A \cup \{g(E_A)\} & \text{if } E_A \neq \emptyset \\ A & \text{if } E_A = \emptyset \end{cases}$ and we notice that A is a maximal chain in X if and only if $f(A) = A$. We define a subset $\mathcal{I} \subseteq \text{Chains}(X)$ to be a tower if

1. $\emptyset \in \mathcal{I}$,
2. $A \in \mathcal{I} \rightarrow f(A) \in \mathcal{I}$,
3. If $\mathcal{C} \subseteq \mathcal{I}$ is a chain under inclusion $\subseteq$, then $\bigcup \mathcal{C} \in \mathcal{I}$.

Towers exist, because $\text{Chains}(X)$ is a tower. We define $\mathcal{I}_0$ to be the smallest tower (i.e. the intersection of all towers). We define an element $A \in \mathcal{I}_0$ to be comparable if $\forall B \in \mathcal{I}_0 A \subseteq B \vee B \subseteq A$. We fix a comparable element $A \in \mathcal{I}_0$ (such elements exist, because $\emptyset$ is comparable) and define $\mathcal{A}_A := \{B \in \mathcal{I}_0 \mid f(A) \subseteq B \vee B \subseteq A\}$. Then $\mathcal{A}_A$ is a tower, so $\mathcal{A}_A = \mathcal{I}_0$. We define $\mathcal{B} := \{A \in \mathcal{I}_0 \mid A \text{ is comparable}\}$. One glance reveals that $\mathcal{B}$ is a tower too. Since every element of $\mathcal{I}_0$ is comparable, then $\mathcal{I}_0$ is a chain and $f(\bigcup \mathcal{I}_0) \in \mathcal{I}_0$ (by definition of towers), meaning that $\bigcup \mathcal{I}_0$ is a maximal chain. **Q.E.D.**

Exercise 7.26. Verify that in the proof of Theorem 7.5.2, $\mathcal{A}_A$ is a tower and that $\mathcal{B}$ is a tower (Hint: For proving that $\mathcal{B}$ is a tower, use the fact that $\mathcal{A}_A = \mathcal{I}_0$).

Theorem 7.5.3 (Kuratowski 1922, Zorn 1935). *Let X be nonempty and $(X, \leq)$ be a partially ordered set such that for any chain $\mathcal{C} \subseteq X$ there exists an upper bound $x \in X$. Then there exists a maximal $x \in X$.*

The theorem is usually referred to as “Zorn’s Lemma” (or “Kuratowski-Zorn Lemma”) and denoted by **ZL**.

Proof. Such an element is an upper bound of any maximal chain in X . **Q.E.D.**

Theorem 7.5.4 (Zermelo). *Let X be a set. Then there exists a well-ordering $\leq$ of X .*

The theorem is usually referred to as “well-ordering theorem” and denoted by **WO**.

Proof. Let $\mathcal{A} := \{R \subseteq X \times X \mid R \text{ is a well-ordering of } \text{dom } R\}$ and for $R, S \in \mathcal{A}$, define $R \leq S$ if $(\text{dom } R, R)$ is an initial segment of $(\text{dom } S, S)$. Then $(\mathcal{A}, \leq)$ is a nonempty partially-ordered set, since $\emptyset \in \mathcal{A}$. For any chain $\mathcal{C} \subseteq \mathcal{A}$, $\bigcup \mathcal{C} \in \mathcal{A}$ is an upper bound of $\mathcal{C}$. By Theorem 7.5.3, $\mathcal{A}$ has a maximal element R . If $\text{dom } R \neq X$, then we can pick any $x \in X \setminus \text{dom } R$ and extend R by putting x after every element of $\text{dom } R$, a contradiction with maximality of R , therefore $\text{dom } R = X$. **Q.E.D.**

Theorem 7.5.5. *In ZF^- , $\mathbf{WO} \rightarrow \mathbf{AC}$.*

Proof. Let $\mathcal{A}$ be a set of nonempty sets. Then there is a well-ordering of $\bigcup \mathcal{A}$. The function $f : \mathcal{A} \rightarrow \bigcup \mathcal{A}$ that maps $A \in \mathcal{A}$ to the smallest element of $A \subseteq \bigcup \mathcal{A}$ is a choice function. **Q.E.D.**

Remark 7.5.6. We’ve proved that in ZF^- :

- $\mathbf{AC} \rightarrow \mathbf{HP}$
- $\mathbf{HP} \rightarrow \mathbf{ZL}$
- $\mathbf{ZL} \rightarrow \mathbf{WO}$
- $\mathbf{WO} \rightarrow \mathbf{AC}$

thus in ZF^- , **AC**, **HP**, **ZL** and **WO** are pairwise equivalent.

7.6 Cardinal Numbers

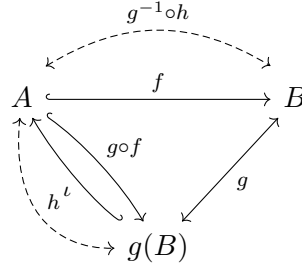
Theorem 7.6.1 (Schröder-Bernstein, Bernstein 1897). *Let A and B be two sets and $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections. Then there exists a bijection $h : A \rightarrow B$.*

This theorem was stated by Cantor in 1887 without proof, and proved by Dedekind in 1887 (unpublished). Cantor (1895) and Schröder (1896) published arguments that were later shown to be incomplete or flawed. Felix Bernstein provided the first correct proof in 1897 (printed by Borel in 1898) independent of **AC**. Schröder then published a flawed proof in 1898, had its errors exposed by Korselt in 1902, confirmed the mistake, and died.

Proof (König 1906). We first consider the case when $B \subseteq A$. Define $\begin{cases} A_0 := A \setminus B, \\ A_{n+1} := f(A_n) \end{cases}$,

$C := \bigcup_{n \in \mathbb{N}} A_n$ and $h(x) := \begin{cases} f(x) & \text{if } x \in C, \\ x & \text{if } x \notin C \end{cases}$. Of course, $h : A \rightarrow B$. Breaking down $h(x) = h(y)$ into cases where both belong to C , neither does, or exactly one does shows that h is an injection. It is just as simple to show that $B \setminus C \subseteq \text{ran } h$ and $B \cap C \subseteq \text{ran } h$, establishing surjectivity and finishing this case.

For $B \not\subseteq A$, we consider the following diagram:



Since g is injective, it is bijective onto its range $g(B) \subseteq A$. Applying the previous case to A and $g(B)$ with injections $g \circ f$ and inclusion ι yields a bijection $h : A \rightarrow g(B)$. Thus $g^{-1} \circ h : A \rightarrow B$ is a bijection. **Q.E.D.**

Definition 7.6.2. For any set A , we define $|A|$ to be the smallest ordinal α such that there exists a bijection $f : A \rightarrow \alpha$.

Theorem 7.6.3. In ZFC^- , for any A , $|A|$ exists.

Proof. There exists a well-ordering $\leq$ of A , so $(A, \leq) \cong_{\text{Ord}} (\alpha, \in)$ for some α , but an order-isomorphism is a bijection. **Q.E.D.**

Definition 7.6.4. An ordinal number κ is a cardinal number if $\kappa = |\kappa|$.

Definition 7.6.5. For two cardinal numbers κ, λ , we define $\kappa \leq \lambda$ if there exists an injection $f : \kappa \rightarrow \lambda$.

Exercise 7.27. Verify that Theorem 7.6.1 may be restated as $|A| \leq |B| \wedge |B| \leq |A| \rightarrow |A| = |B|$.

Theorem 7.6.6 (Cantor). For any A , $|A| < |\mathcal{P}(A)|$.

Proof. It suffices to show that there exists no surjection $f : A \rightarrow \mathcal{P}(A)$. Suppose that it exists and define a set $C := \{x \in A \mid x \notin f(x)\}$. Then $C = f(x)$ for some x , so $x \in C \iff x \notin f(x) \iff x \notin C$, a contradiction. **Q.E.D.**

Definition 7.6.7. Let κ and λ be cardinal numbers and let A, B be sets such that $|A| = \kappa, |B| = \lambda$. We define:

- $\kappa + \lambda := |A \sqcup B|$
- $\kappa \cdot \lambda := |A \times B|$
- $\kappa^\lambda := |A^B|$

Chapter 8

Theory of Computation

8.1 Decidability and Semidecidability

We conclude by studying the limits of computation: what mathematical problems can computers solve, and what is unsolvable?

Definition 8.1.1. A set $A \subseteq \mathbb{N}$ is *decidable* (or *computable*) if there exists an algorithm that, on input $n \in \mathbb{N}$, returns **Yes** if $n \in A$, and **No** if $n \notin A$.

Definition 8.1.2. A set $A \subseteq \mathbb{N}$ is *semidecidable* (or *recursively enumerable*) if there exists an algorithm that, on input $n \in \mathbb{N}$, returns **Yes** if $n \in A$, and runs forever (or halts without output) if $n \notin A$.

Remark 8.1.3. When analyzing algorithms, we consider an idealized model of computation (such as a Turing Machine) with infinite memory and no physical processing limitations.

Exercise 8.1. Let $A, B \subseteq \mathbb{N}$ be decidable. Prove that A^c , $A \cup B$, and $A \cap B$ are decidable.

Exercise 8.2. Let $A, B \subseteq \mathbb{N}$ be semidecidable. Prove that $A \cup B$ and $A \cap B$ are semidecidable.

Definition 8.1.4. An algorithm *enumerates* A if it lists the elements of A such that every element of A is eventually listed in finite time, and no elements outside of A are ever listed.

Remark 8.1.5. An algorithm enumerating A may print the same element multiple times.

Example 8.1.6. The following C program enumerates all multiples of 2 and 3 under 20:

```
#include <stdio.h>
int main()
{
    int i, j;
    for(i = j = 0; i <= 20 && j <= 20; i += 2, j += 3){
```

```

        printf("%d %d ", i, j);
    }
    return 0;
}

```

Theorem 8.1.7. *A set A is semidecidable if and only if it can be enumerated by an algorithm.*

Proof. ($\implies$) Let X be an algorithm that semidecides A . We design an enumerator for A by executing X on inputs in a grid-like fashion (dovetailing) to avoid getting stuck on infinite paths:

```

num = 0, b = 0
while True:
    while num <= b:
        Run X on num for b steps.
        If X halted and said Yes, print num.
        num++
    num = 0, b++

```

($\impliedby$) Let Y be an enumerator for A . We semidecide A on input a by running Y and halting with **Yes** if a is ever printed. ***Q.E.D.***

Remark 8.1.8 (The Trap of Naive Enumeration). It is highly instructive to analyze why the nested loop design in the proof above is necessary. A naive approach to enumerating A would be to run our semidecision algorithm X on 0 until it halts, then run X on 1 until it halts, and so on.

However, if $0 \notin A$, the algorithm $X(0)$ will run forever. Our naive enumerator would get permanently stuck on the very first input, and we would never discover that $1 \in A$ (even if $X(1)$ would have halted in just two steps). The grid-allocation method (historically known as *dovetailing*) ensures that we execute a bounded number of steps on multiple computations, preventing any single infinite loop from blocking our progress.

Theorem 8.1.9 (Kleene). *A set $A \subseteq \mathbb{N}$ is decidable if and only if both A and A^c are semidecidable.*

Exercise 8.3. Prove Theorem 8.1.9.